

AI and Storyboarding for Game Design

Graduate course · AI and AI engineering for game design and development

Four questions, in order

01	**Game design**	What are we designing?	WHAT · WHY · HOW
02	**Storyboarding**	How do we plan time and intent before it gets expensive?	WHAT · WHY · HOW
03	**Storyboarding for game design**	What changes when the viewer has agency?	WHAT · WHY · HOW
04	**How AI helps**	Which layers can AI take on, and which need a human?	WHAT · WHY · HOW

THE ONE IDEA

In film, a storyboard shows what the camera sees.

In games, it shows what the player can do and feel.

BEFORE WE START

Two books for the habits behind good work



Steal Like an Artist

AUSTIN KLEON · 2012 · 10 CHAPTERS COURSE

01 **Steal like an artist.**
> Take the structure of an exemplar, not its surface.

02 Don’t wait until you know who you are to get started.

03 Write the book you want to read.

04 **Use your hands.**
> Thumbnail on paper before any tool.

05 Side projects and hobbies are important.

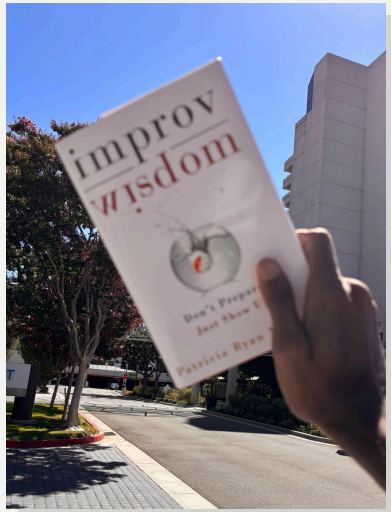
06 **The secret: do good work and share it with people.**
> Show boards early. Critique is the point.

07 Geography is no longer our master.

08 Be nice. The world is a small town.

09 Be boring. It’s the only way to get work done.

10 **Creativity is subtraction.**
> Cut panels until each one carries a decision.



Improv Wisdom

PATRICIA RYAN MADSON · 2005 · 13 MAXIMS

01 **Say yes.**
> Build on a teammate’s idea before judging it.

02 Don’t prepare.

03 Just show up.

04 **Start anywhere.**
> Any panel can be the first one.

05 **Be average.**
> A rough thumbnail people change beats a polished one they don’t.

06 Pay attention.

07 Face the facts.

08 Stay on course.

09 Wake up to the gifts.

10 **Make mistakes, please.**
> Playtests exist to find what breaks.

11 Act now.

12 Take care of each other.

13 Enjoy the ride.

Both apply beyond games: to work, to teams and to life.

01

Game design

What are we designing?

An experience, built indirectly through rules.



A Pac-Man arcade cabinet.

YosemiteYamper · CC BY-SA 4.0 · Wikimedia Commons

The experience is the product

Players do not want the game itself; they want the experience it creates. The game is the machine that makes it.

FOR BOARDS

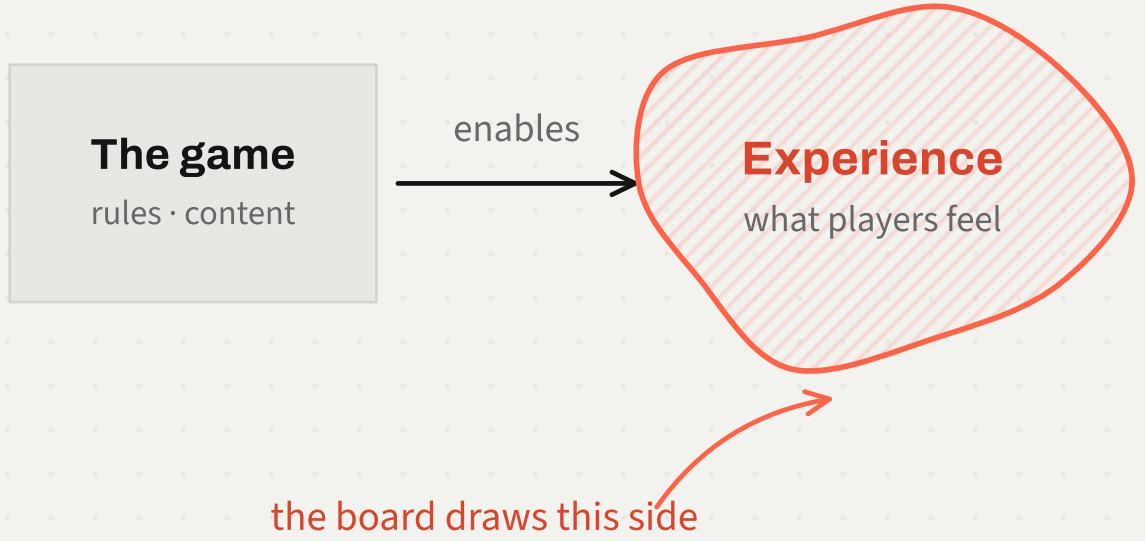
A game storyboard should draw the experience, not just the content.

NUANCE

The same content can produce very different experiences, depending on how it is paced and played.

EXAMPLE

Journey's rules are simple. The awe and companionship it creates are the point.



ASK THE ROOM

Name a game you love. What did it make you feel, and which rule made you feel it?

SOURCE Schell, The Art of Game Design: A Book of Lenses (3rd ed., 2019)

Design is second-order

Designers write rules, not play. Play emerges when people use the rules.

FOR BOARDS

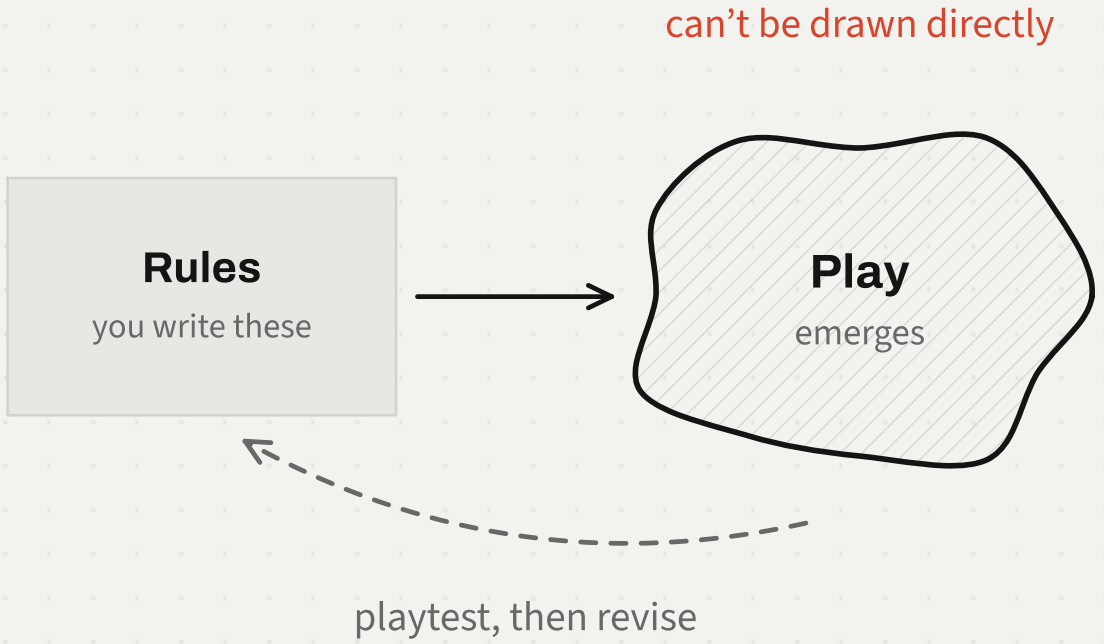
You cannot storyboard play directly. You storyboard the conditions that make the play you want likely.

NUANCE

That is why boards need playtests: the emergent part only appears in play.

EXAMPLE

The rules of chess fit on one page. The games played with them do not.



ASK THE ROOM

What is one thing players did in a game that its designers probably never planned?

SOURCE Salen & Zimmerman, Rules of Play: Game Design Fundamentals (2003)

Mechanics, dynamics, aesthetics



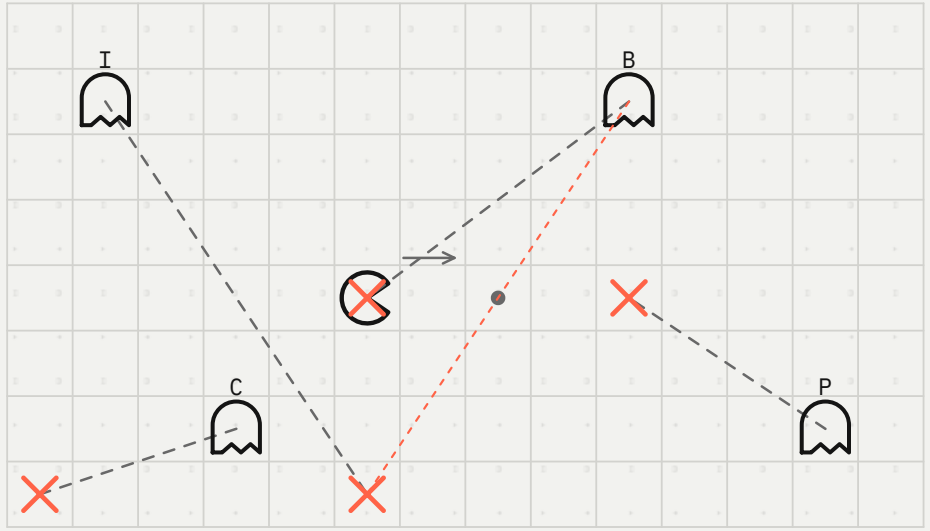
Designers set the mechanics. Play produces the dynamics. Players feel the aesthetics.

“Aesthetics” means feelings, not visual style. Pac-Man: ghost targeting rules → chases and near misses → tension.

SOURCE Hunicke, LeBlanc & Zubek, MDA: A Formal Approach to Game Design and Game Research (2004)

CASE STUDY · PAC-MAN · NAMCO · 1980

Pac-Man’s four ghosts



Blinky
targets your tile

Pinky
targets four tiles ahead of you

Inky
doubles the line from Blinky to two tiles ahead of you

Clyde
chases when far; retreats to his corner when close

target tile

WHAT THEY DID

Each ghost picks its target with its own rule. Blinky targets your tile. Pinky aims four tiles ahead of you. Inky doubles the line from Blinky to the tile two ahead of you. Clyde chases when far away and retreats to his corner when close.

WHAT IT TEACHES

Four small rule differences produce four readable personalities. The chase becomes something players can learn to play against: MDA in one screen.

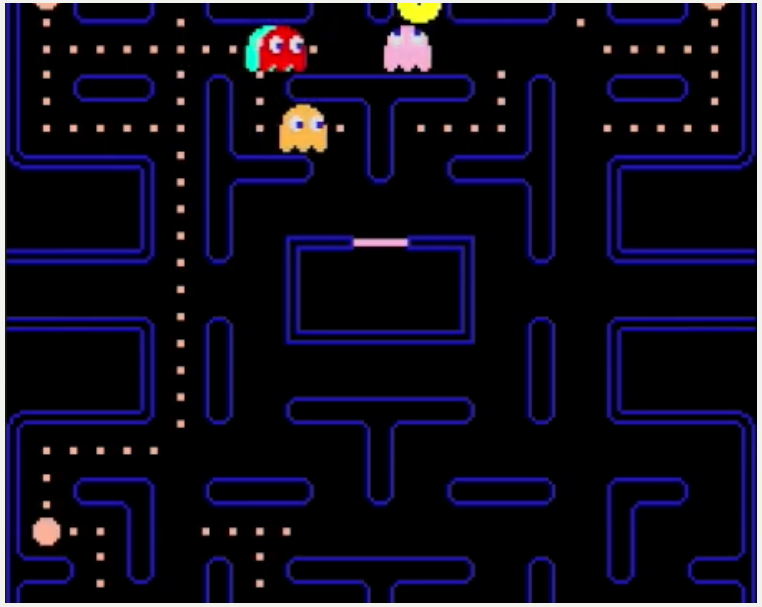
DISCUSS

What would happen to the feeling if all four ghosts simply chased you?

SOURCE Pittman, The Pac-Man Dossier

LOOK CLOSER

Pac-Man: the whole system on one screen



Pac-Man (1980). Four ghosts, one maze, every rule on one screen.

Bandai Namco Entertainment America · CC BY 3.0 · Wikimedia Commons



A Pac-Man arcade cabinet.

YosemiteYamper · CC BY-SA 4.0 · Wikimedia Commons



Toru Iwatani, creator of Pac-Man, at GDC 2011.

Official GDC · CC BY 2.0 · Wikimedia Commons

ASK THE ROOM

Which rules can you see on this screen, and which are hidden?

Fun is learning

Fun is the brain mastering new patterns. Once a pattern is mastered, it turns boring.

FOR BOARDS

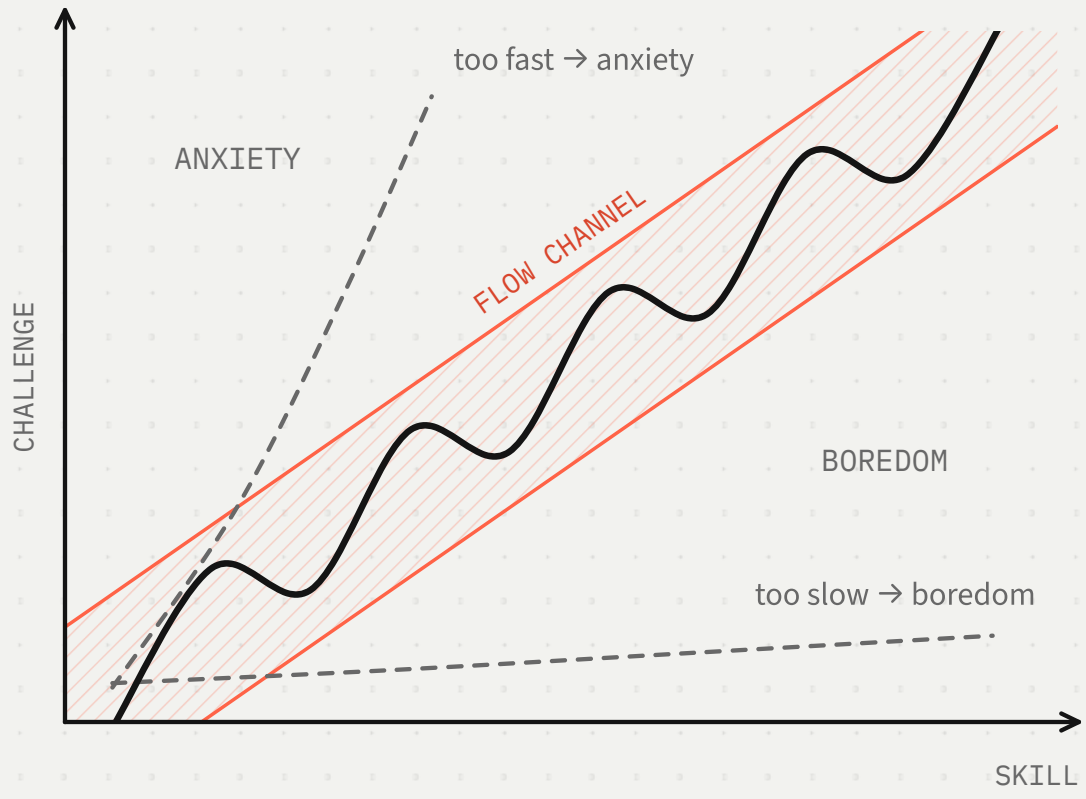
Pacing is the rate at which new patterns arrive, so a storyboard is also a teaching plan.

NUANCE

Too much challenge for the player's skill feels anxious; too little feels boring. The space between is the flow channel.

EXAMPLE

A Mario level introduces one idea, then varies it until the player has mastered it.



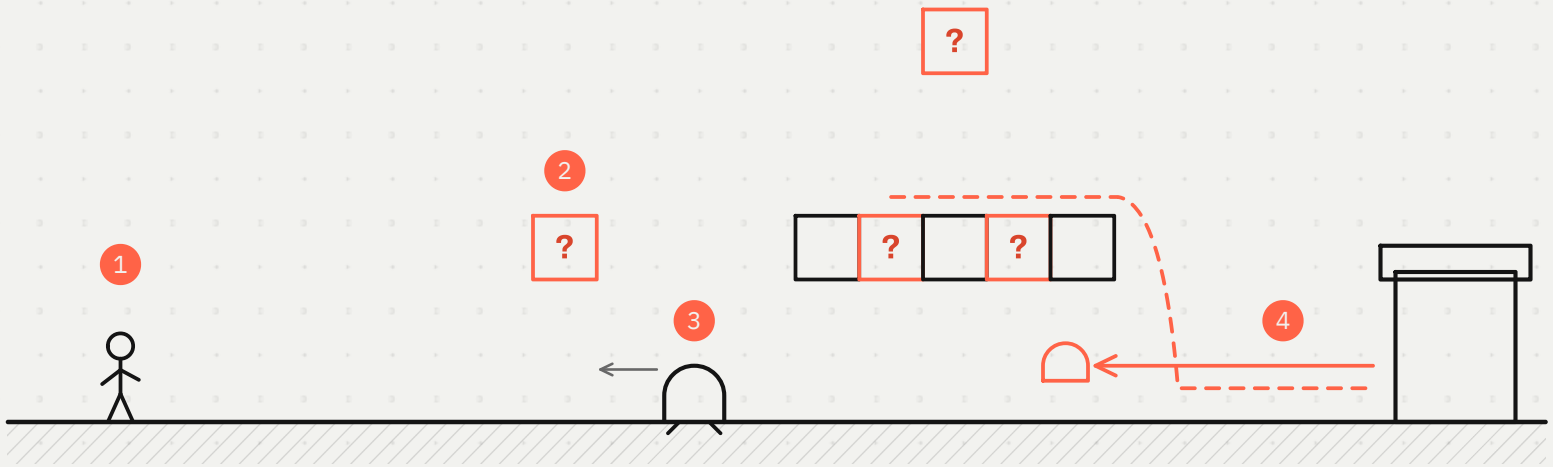
ASK THE ROOM

Where in your favourite game did you stop learning, and what did the game do next?

SOURCE Koster, A Theory of Fun for Game Design (2nd ed., 2013) · Csikszentmihalyi, Flow: The Psychology of Optimal Experience (1990)

CASE STUDY · SUPER MARIO BROS. · NINTENDO · 1985

World 1-1 teaches without words



schematic, not to scale

1. Start on the left, open space ahead
2. A lone ? block invites a bump
3. The first enemy is easy to jump on
4. The mushroom rolls away, hits the pipe, comes back

WHAT THEY DID

The first level was designed so players learn by playing. The Goomba was created as an easier first enemy than the planned Koopa. The first mushroom rolls away, bounces off a pipe and comes back to the player.

WHAT IT TEACHES

The first minute of a level is a lesson plan, and it can be boarded as one.

DISCUSS

Where in your game does the first minute teach the core verb, without any text?

SOURCE World 1-1 (Wikipedia), drawing on Miyamoto and Tezuka’s 2015 Eurogamer interview

Playcentric iteration

Prototype, playtest, evaluate, revise: early and often, with players in the loop from the start.

FOR BOARDS

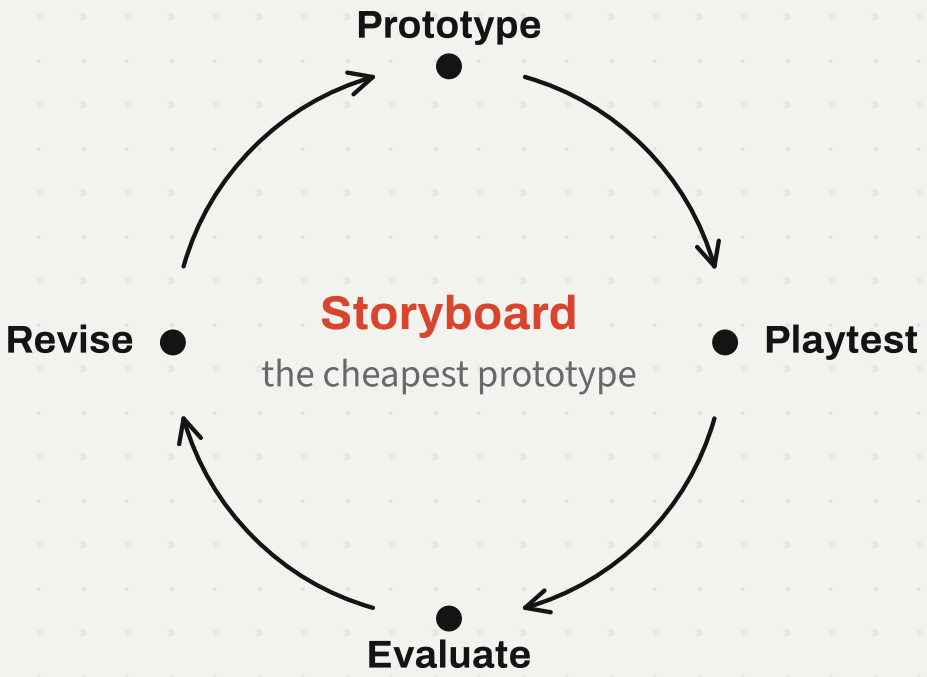
A storyboard is the cheapest prototype. You can “playtest” it by walking someone through it.

NUANCE

Each round needs a clear question, or iteration turns into churn.

EXAMPLE

Paper prototypes are tested before any code is written.



ASK THE ROOM

What is the one question your next prototype must answer?

SOURCE Fullerton, Game Design Workshop

Lenses

A lens is a set of questions that makes you look at a design from one angle.

FOR BOARDS

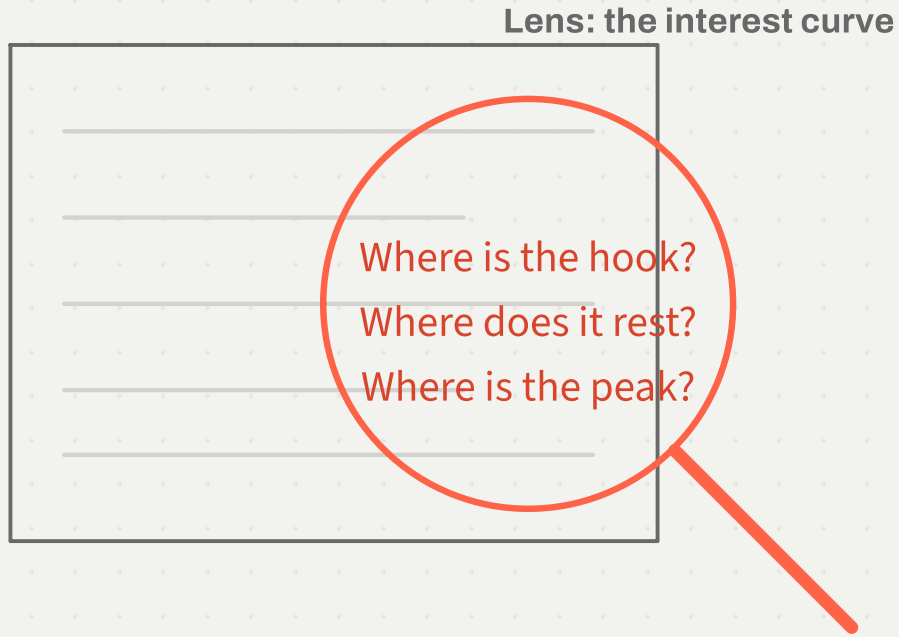
Lenses make storyboard critique concrete: review the board through named questions instead of taste alone.

NUANCE

Lenses prompt thinking. They are not rules to satisfy.

EXAMPLE

The interest curve lens asks how a player's interest rises and falls over time.



ASK THE ROOM

Which single lens would change your current design most if you applied it honestly?

SOURCE Schell, The Art of Game Design: A Book of Lenses (3rd ed., 2019)

Myth vs reality

MYTH

~~In MDA, aesthetics means the art style.~~

REALITY

It means what the player feels: challenge, discovery, fellowship and so on.

MYTH

~~If the rules are right, the experience follows.~~

REALITY

Rules only make some experiences likely. Playtests show which ones you actually got.

TRY THIS · 15 MIN

MDA teardown

1. Pick a game you know well.
2. Write three chains: mechanic → dynamic → aesthetic.
3. Find one aesthetic that nothing produces, or one mechanic that serves no aesthetic.

OUTPUT

One page: three chains and one gap.

INQUIRY

Your team wants a co-op game that feels like fellowship.

Q1 Which dynamic would produce fellowship?

Q2 Which mechanic forces that dynamic, instead of just allowing it?

Q3 How would a playtest tell you that you got fellowship and not frustration?

Check yourself

1. **Why does MDA say designers and players meet a game from opposite ends?**

   Designers can only set the mechanics. Players first meet the aesthetics, what they feel, and only later work out the rules behind it.

2. **What does “aesthetics” mean in MDA?**

   The player’s emotional response, such as challenge or discovery. Not the visual style.

3. **Why is game design “second-order”, and what does that mean for a storyboard?**

   Designers write rules and play emerges from them. A board can only plan the conditions for the play you want, so it has to be playtested.

02

Storyboarding

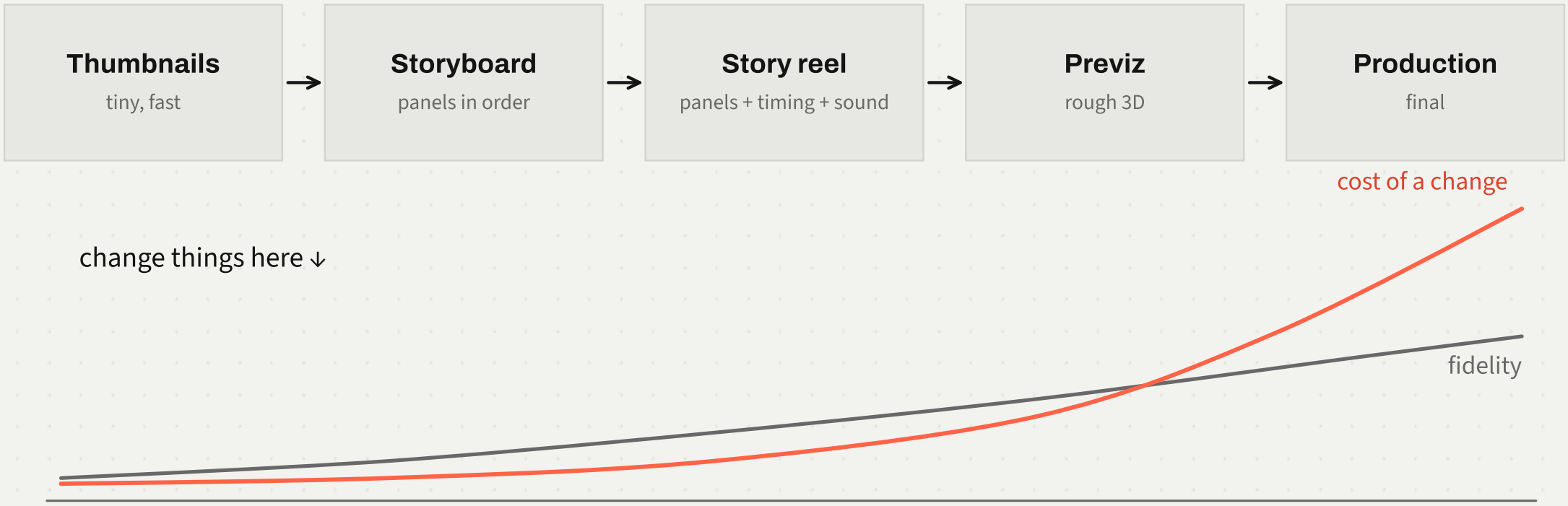
How do we plan time and intent before it gets expensive?

Draw it in order, roughly, and argue about it.



Storyboard and concept frames for Charge, a Blender Studio short film.
Andy Goralczyk / Blender Foundation · CC BY 4.0 · Wikimedia Commons

What a storyboard is



A sequence of drawings that plans what happens, in what order and how it is framed, before anything expensive is made.

Fidelity rises along the pipeline, and so does the cost of a change.

SOURCE Catmull with Wallace, Creativity, Inc. (2014)

LOOK CLOSER

Storyboards in production



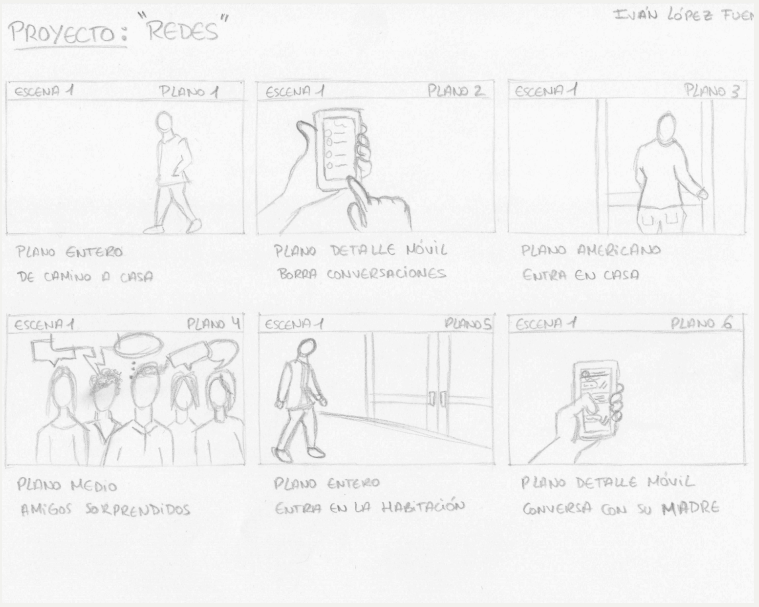
Storyboard and concept frames for Charge, a Blender Studio short film.

Andy Goralczyk / Blender Foundation · CC BY 4.0 · Wikimedia Commons



Storyboard frames for Spring, a Blender Studio short film.

David Revoy / Blender Foundation · CC BY 4.0 · Wikimedia Commons



A hand-drawn storyboard sheet with numbered scenes and shots.

Iván López Fuente · CC BY-SA 4.0 · Wikimedia Commons

ASK THE ROOM

What does each board leave out, and why is that useful?

Fail cheaply

Boards and reels let a team find what does not work while a change costs a pencil stroke.

FOR BOARDS

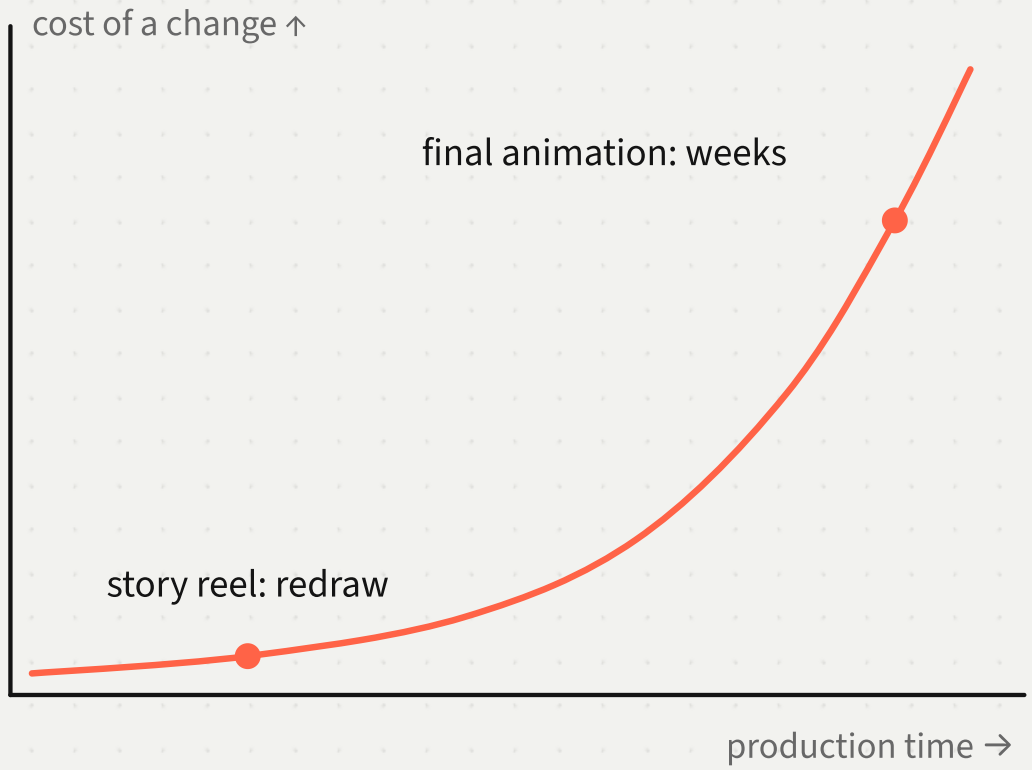
The value of a board comes from being cheap to change.

NUANCE

Catmull says Pixar's films are bad early on. The process, including candid notes from the Braintrust, exists to make them better.

EXAMPLE

A story problem caught in a reel costs a redraw. Caught in final animation, it costs weeks.



ASK THE ROOM

What was the most expensive late change on a project you know, and where could a board have caught it?

SOURCE Catmull with Wallace, Creativity, Inc. (2014)

CASE STUDY · FEATURE FILMS · PIXAR · ONGOING

Pixar’s story reels

WHAT THEY DID

Storyboards are cut together with temporary voices and music into story reels and screened for the Braintrust, a group of directors who give candid notes. Reels are reworked many times before animation starts.

WHAT IT TEACHES

A board’s job is to be watched, judged and changed while change is still cheap.

DISCUSS

What would a story reel be for your game, and who is your Braintrust?



Ed Catmull, co-founder of Pixar, in 2014.

Steve Jurvetson · CC BY 2.0 · Wikimedia Commons

SOURCE Catmull with Wallace, Creativity, Inc. (2014)

Align the team

A board turns one person’s intent into something everyone can see, point at and argue about.

FOR BOARDS

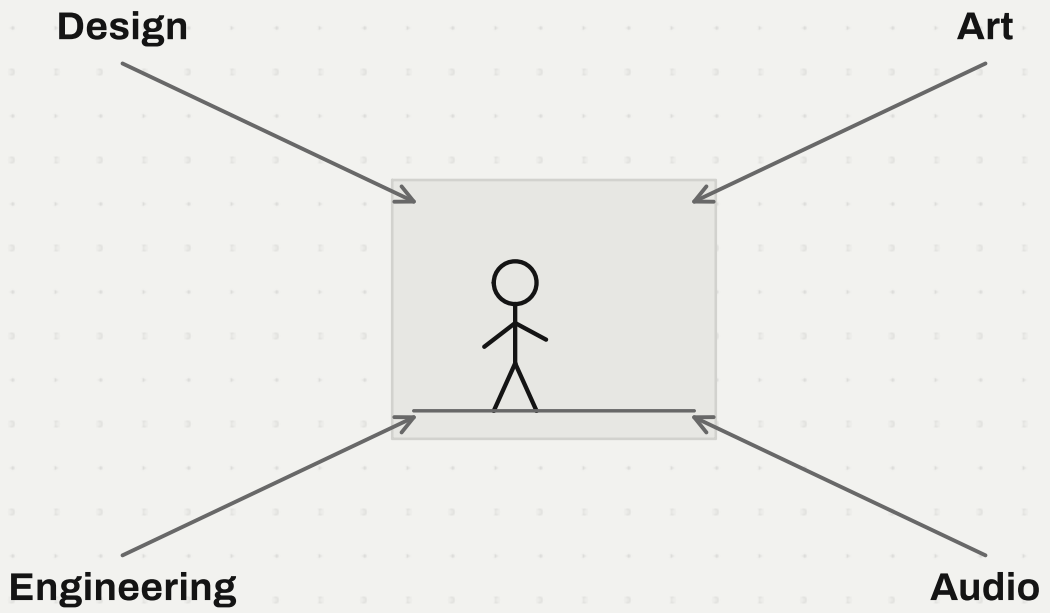
For games, the board is where design, art, engineering and audio agree on the same moment.

NUANCE

A board that nobody argues with is usually too polished or too vague.

EXAMPLE

A director, an animator and a level designer can disagree about panel 3 instead of about a paragraph.



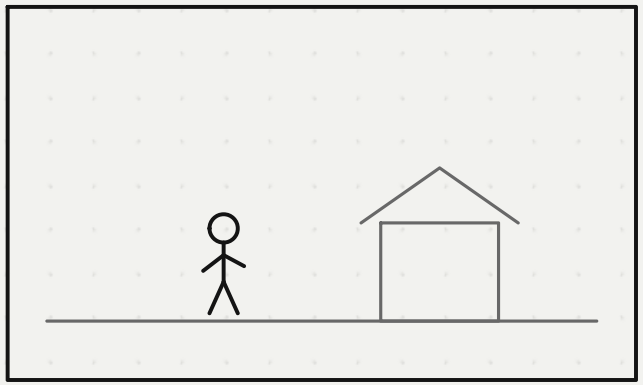
one moment everyone can point at

ASK THE ROOM

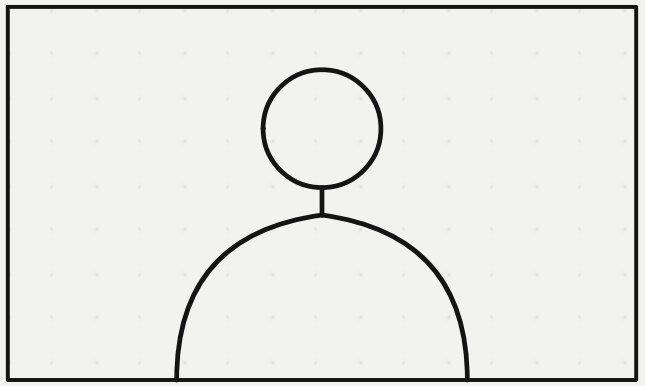
If five people read your board, would they all describe the same moment?

SOURCE Catmull with Wallace, Creativity, Inc. (2014) · Buxton, Sketching User Experiences (2007)

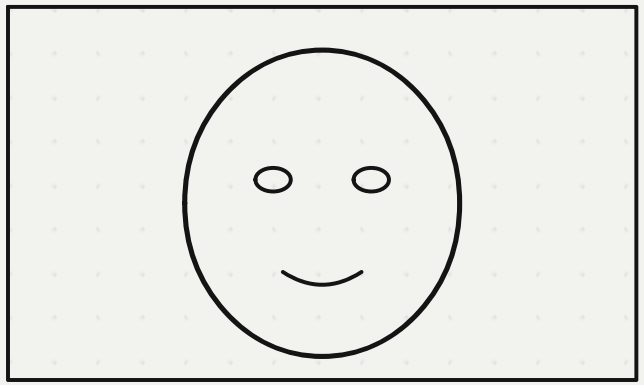
Shot grammar



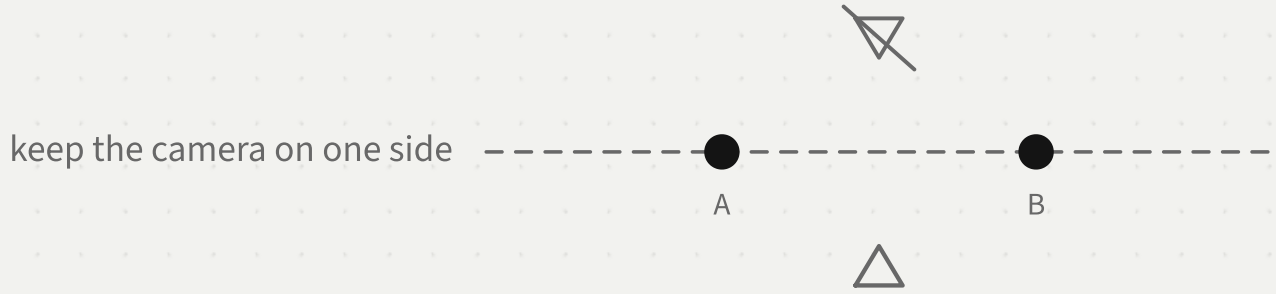
Wide



Medium



Close-up



in games, the player may hold this camera

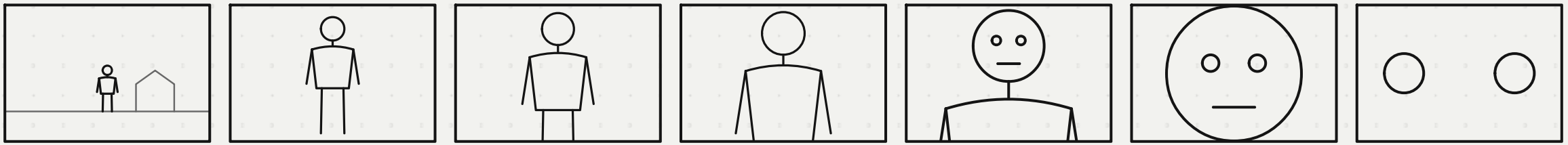
Shot size, angle, movement and continuity decide what the viewer sees and understands.

All of it assumes the director holds the camera.

SOURCE Katz, Film Directing Shot by Shot (1991)

STANDARD

Shot sizes



ELS
extreme long

LS
long

MLS
medium long

MS
medium

MCU
medium close-up

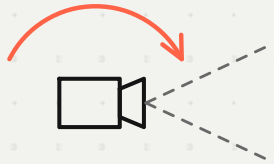
CU
close-up

ECU
extreme close-up

STANDARD

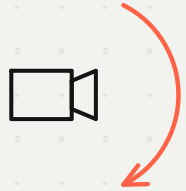
Camera moves

Pan



turns left or right
on the spot

Tilt



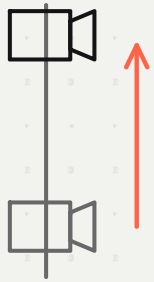
turns up or down
on the spot

Track / dolly



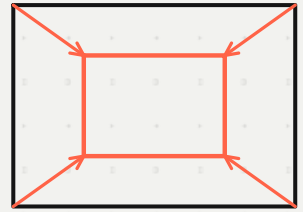
the camera itself
travels

Crane



the camera rises
or lowers

Zoom



the lens changes;
the camera stays

Notation: arrows inside the frame show subject movement. A frame drawn inside the frame, with arrows between the corners, shows a zoom or push-in.

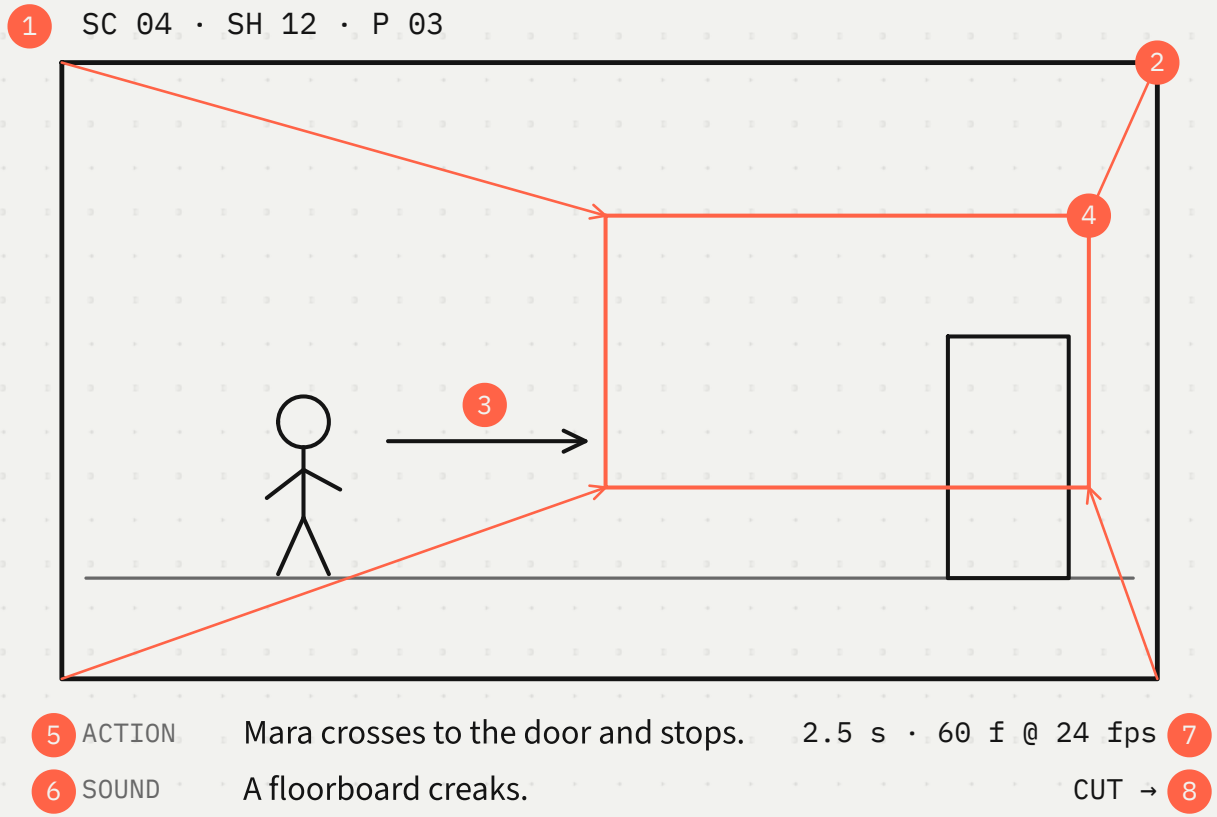
STANDARD

Continuity rules

180° rule	Keep the camera on one side of the line between two subjects, so left and right stay consistent.
30° rule	Between two shots of the same subject, move the camera at least 30°, or the cut looks like a jump.
Screen direction	Keep travel in a consistent direction across cuts, for example always left to right on a journey.
Eyeline match	Cut from a look to what is being looked at, and keep the angle believable.
Match on action	Cut in the middle of a movement so it continues across the cut.

STANDARD

Anatomy of a storyboard panel



1. ID: scene · shot · panel
2. Frame: what the camera sees (16:9)
3. Subject movement: arrow in the frame
4. Camera move: end frame + corner arrows
5. Action: one line
6. Dialogue and sound
7. Timing: seconds or frames
8. Transition: cut, dissolve, fade

Contrast and affinity

Visual contrast (in space, line, shape, tone, colour, movement and rhythm) raises intensity. Affinity, or similarity, lowers it.

FOR BOARDS

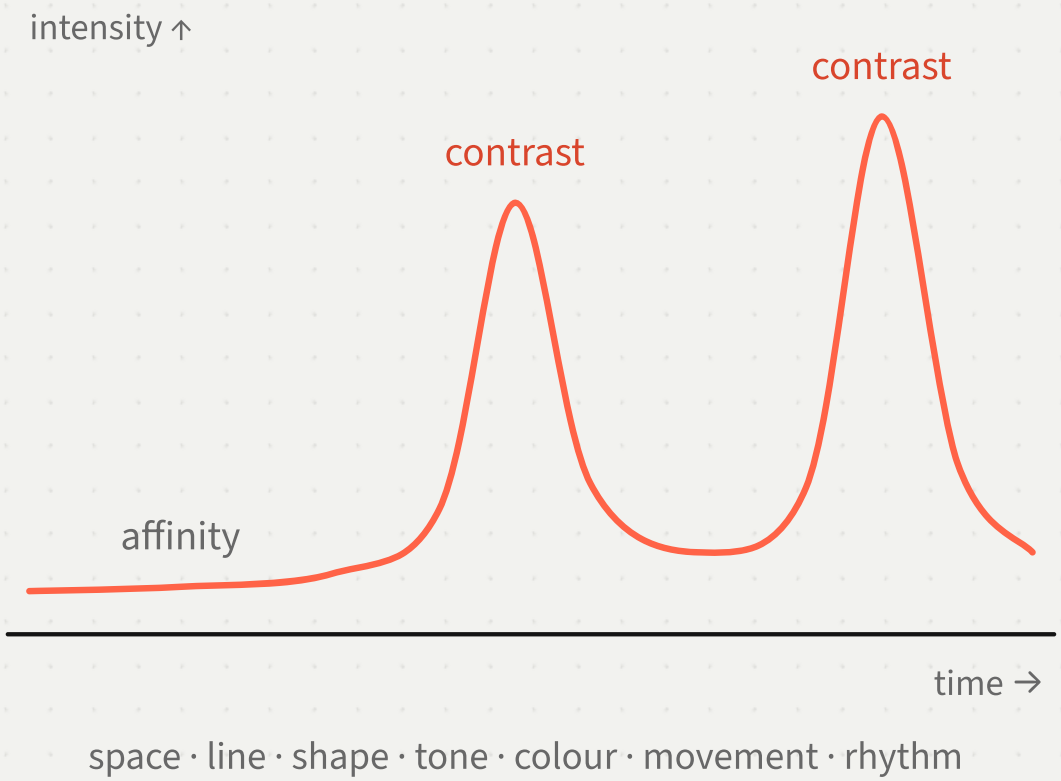
Intensity can be planned and drawn, not only felt.

NUANCE

Structure comes from changes in intensity over time, not from constant intensity.

EXAMPLE

Calm, similar shapes and colours, then a sharp break when the threat appears.



ASK THE ROOM

Where is the biggest visual contrast in your level, and is it where the biggest moment is?

SOURCE Block, The Visual Story

Closure: six panel transitions

Readers fill in what happens between panels, in the gutter.

TRANSITION	WHAT THE READER FILLS IN
Moment-to-moment	Very little time passes: an eye closing, frame by frame.
Action-to-action	One subject, one action after another: the swing, then the hit.
Subject-to-subject	Same scene, different subject: the batter, then the crowd.
Scene-to-scene	A jump across time or place.
Aspect-to-aspect	Different parts of one place or mood; time seems to stand still.
Non-sequitur	No logical link between panels.

In a game, the player fills the gutter with their own action.

SOURCE McCloud, Understanding Comics (1993)

Fidelity is a choice

Rough sketches are quick, cheap, disposable and ambiguous, so they invite critique and alternatives. Polished images signal that decisions are made.

FOR BOARDS

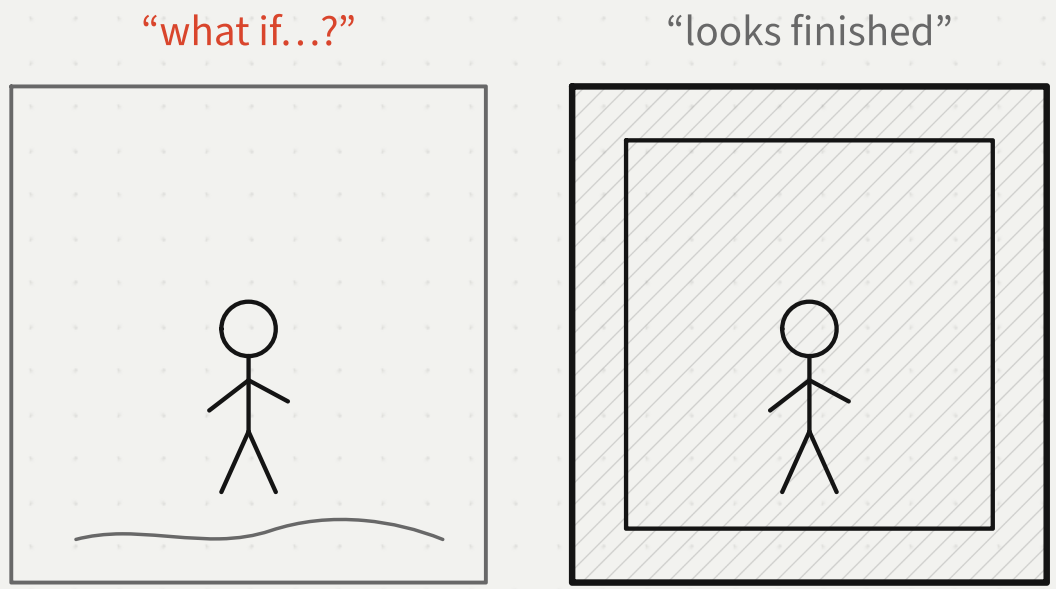
Match fidelity to the question you are asking.

NUANCE

This becomes central once AI makes polish cheap (Module 4).

EXAMPLE

Stick figures for the order of events; finished frames only once the order is settled.



Rough
invites critique

Polished
signals “done”

ASK THE ROOM

What is the lowest fidelity at which your current question can still be answered?

SOURCE Buxton, Sketching User Experiences (2007)

CASE STUDY · SUPER MARIO ODYSSEY · NINTENDO · 2017

Cutscene boards at Nintendo

WHAT THEY DID

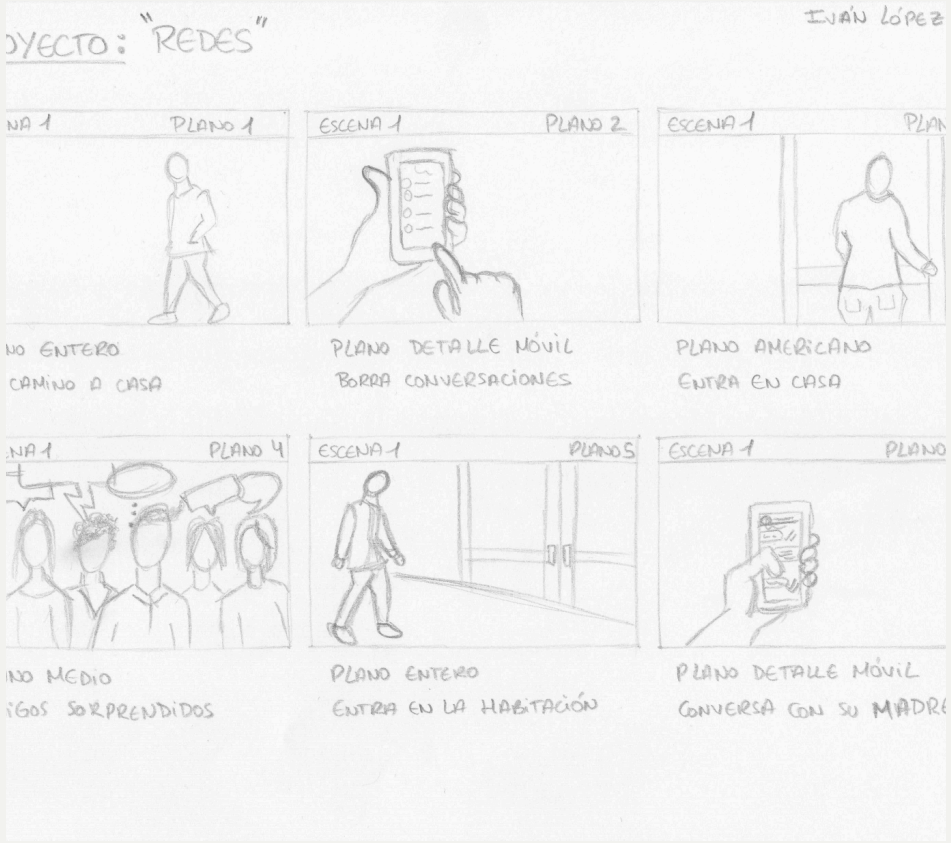
The game’s cutscenes were storyboarded as hand-drawn panels before animation. The boards are published in the game’s art book.

WHAT IT TEACHES

Game cinematics still use film boards. It is gameplay that needs the other layers of the stack.

DISCUSS

Which moments in your game must be authored shots, and which must stay in the player’s hands?



A hand-drawn storyboard sheet with numbered scenes and shots.

Iván López Fuente · CC BY-SA 4.0 · Wikimedia Commons

SOURCE Super Mario Odyssey: cutscene storyboards (Game Anim)

Myth vs reality

MYTH

~~A storyboard needs good drawing.~~

REALITY

It needs clear order, framing and intent. Stick figures do the job.

MYTH

~~The more detailed the board, the better.~~

REALITY

Detail too early slows change and discourages critique.

TRY THIS · 15 MIN

Six thumbnails

1. Board a ten-second moment in six rough thumbnails.
2. Label each transition using McCloud’s six types.
3. Redraw it with two panels removed. Does the reader still follow?

OUTPUT

Two strips and one note on what the gutter now does.

INQUIRY

A lead asks for final-looking boards for a pitch next week.

Q1 What do you gain from polish here?

Q2 What do you lose in critique?

Q3 Could you show two fidelities to two audiences?

Check yourself

1. **Why do studios storyboard at all?**

   To fail cheaply, and to give a team one visible version of the intent before expensive production.

2. **What is closure, and how does it change in games?**

   The reader imagines what happens between panels. In a game, the player fills that gap with their own action.

3. **Why can a rough board be better than a polished one?**

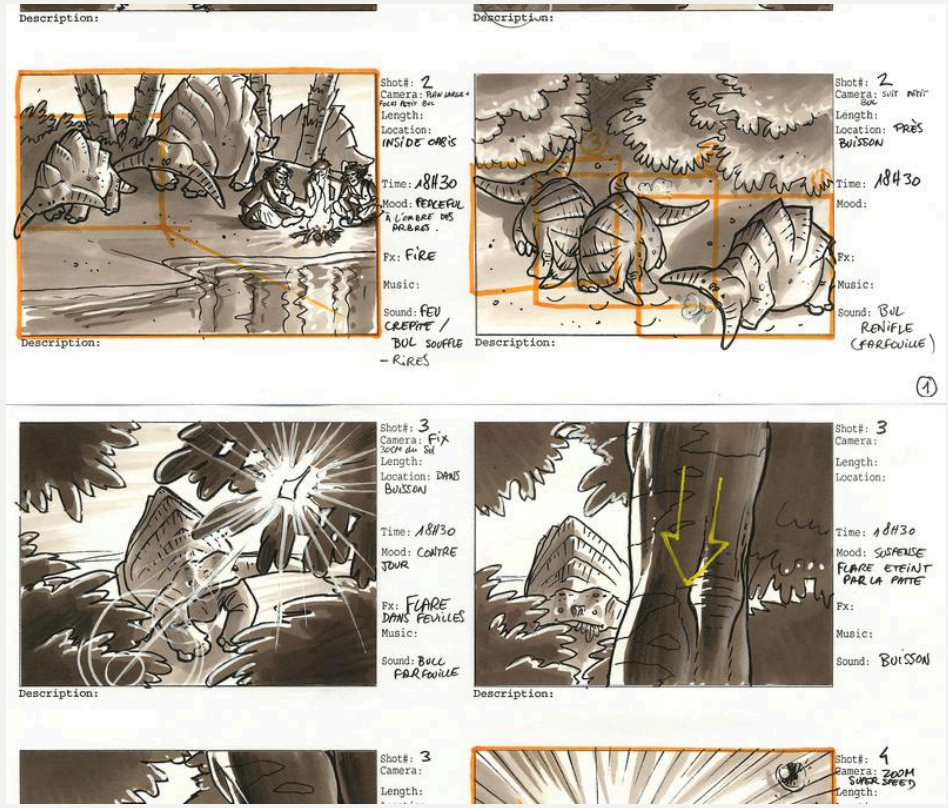
   Rough boards invite critique and alternatives. Polish signals that decisions are final.

03

Storyboarding for game design

What changes when the viewer has agency?

You board what the player can do and feel, at many layers.



Storyboard panels for the online game Ryzom.

Ryzom · CC BY-SA 2.0 · Wikimedia Commons

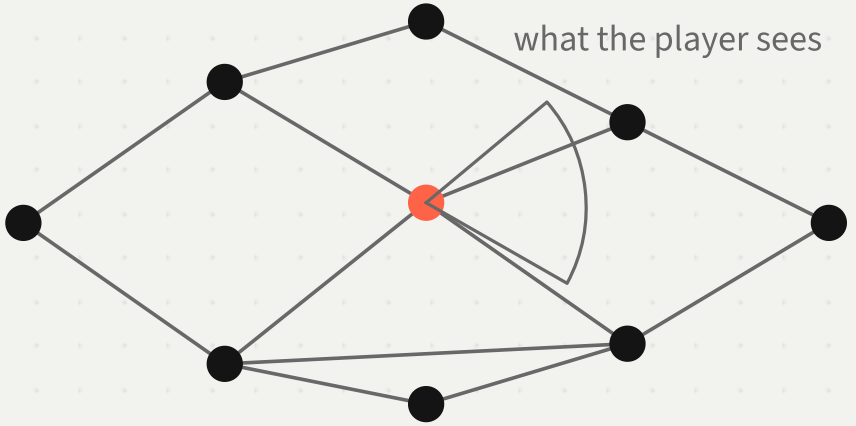
A spec for an interaction

Film



one order · the director’s camera

Game



a graph of states · the player’s camera

A film board shows what the camera sees. A game board shows what the player can do and feel.

Film board vs game board

FILM STORYBOARD	GAME STORYBOARD
A linear sequence of shots	A graph of states, choices and branches
The camera is authored	The camera is often driven by the player
Tells a story	Shows the player’s experience: beats, pacing, feedback loops
One output: shots	Many outputs: cinematics, gameplay beats, UX flows, level beat charts, tutorials
One unit: the shot	Many units: beat, choice, room, panel, shot, screen
One representation: images	Many: text, graph, map, curve, image, video, state machine

Narrative architecture

Game stories are often told through space (Jenkins, 2004).

Evoked

Calls up a story the player already knows.

A game set in a famous film’s world.

Enacted

The player acts the story out through their goals.

A hero’s quest, played step by step.

Embedded

Story clues are placed in the world to be found.

Notes, bodies and broken rooms that tell what happened.

Emergent

Stories arise from systems as people play.

The Sims.

So game boards need maps and paths, not only shots.

SOURCE Jenkins, Game Design as Narrative Architecture (2004)

The storyboarding stack

STRUCTURE

01	Concept	The pitch
02	Narrative structure	The beat
03	Branching structure	The choice

EXPERIENCE

04	Experience pacing	Intensity over time
05	Level and space	The room
06	Gameplay moment	The panel + input

PRODUCTION

07	Cinematic	The shot
08	Interface flow	The screen
09	Validation	The playtest

06

Gameplay moment

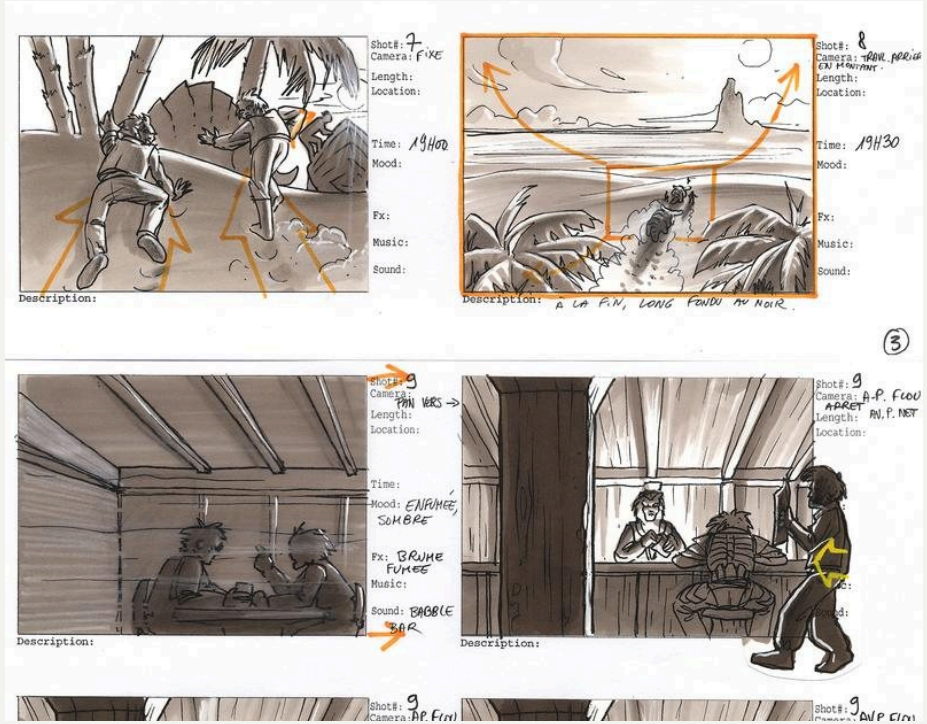
UNIT	The panel + input
METHODS	Comic-style panels annotated with the 3Cs; panel transitions
REFERENCES	McCloud, Block
REPRESENTATION	Image

Select a layer.

SOURCE Rogers, Level Up! The Guide to Great Video Game Design (3rd ed.)

LOOK CLOSER

From board to game: Ryzom



A Ryzom storyboard page: numbered panels with notes beside each.

Ryzom · CC BY-SA 2.0 · Wikimedia Commons



Ryzom in play: what the boards became.

Ryzom · CC BY-SA 2.0 · Wikimedia Commons

ASK THE ROOM

What changed between the board and the shipped screen? Which decisions did the board settle?

Many units, many representations



Each representation needs a different kind of model. Hold that thought for Module 4.

LOOK CLOSER

Which board would you draw first?



SuperTuxKart: the track map sits in the corner of the screen.

QwertyChouskie · CC BY-SA 4.0 · Wikimedia Commons

BOARD FIRST

A map: the track is the level.



Minetest: a sandbox where play comes from systems, not scripts.

Perttu Ahola, Vanessa Ezekowitz et al. · CC BY-SA 3.0 · Wikimedia Commons

BOARD FIRST

A loop: play comes from systems, not scenes.



0 A.D.: a strategy game read as a map of territory and resources.

Wildfire Games · CC BY-SA 3.0 · Wikimedia Commons

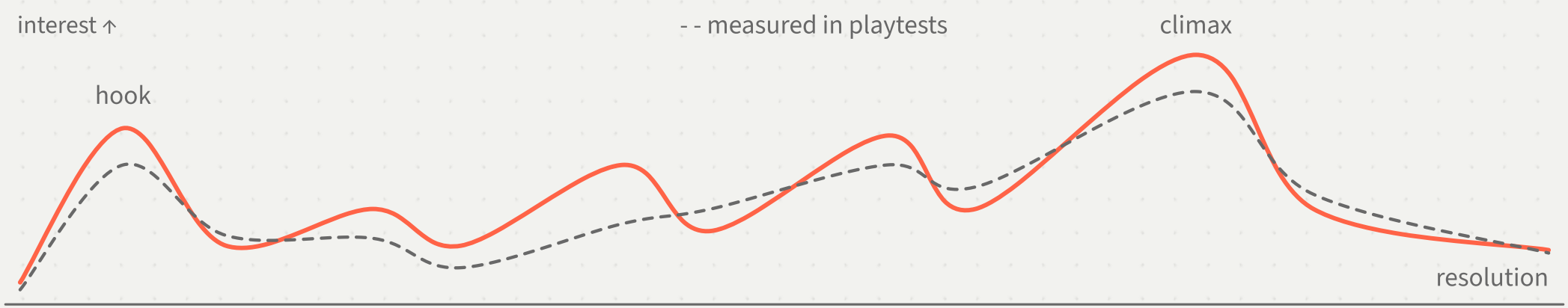
BOARD FIRST

A map and a loop: territory plus an economy.

ASK THE ROOM

Decide before the answer appears. There is more than one defensible choice.

Beat charts and the interest curve



	Level 1	Level 2	Level 3	Level 4	Level 5
New mechanic	run, jump	wall jump	grapple	—	all of them
Threat	slimes	bats	turrets	mixed	the boss
Story beat	leave home	meet the guide	the bridge falls	the guide is lost	return home

sample beat chart

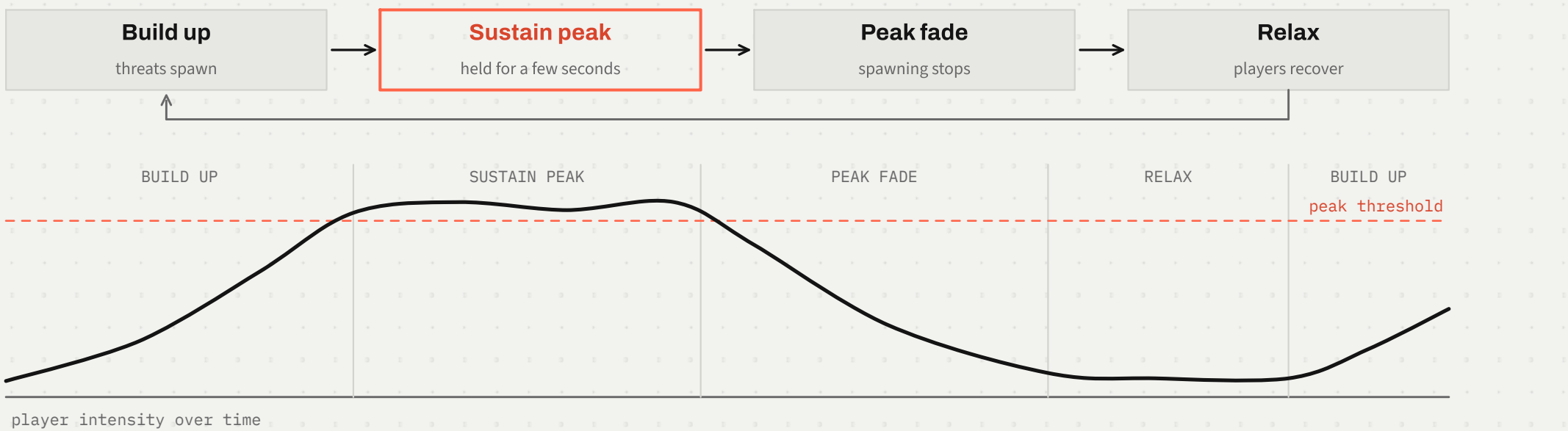
The interest curve: a hook, rising peaks with rests, a climax, a resolution.

A planned curve is a guess. Check it against playtests.

SOURCE Rogers, Level Up! The Guide to Great Video Game Design (3rd ed.) · Schell, The Art of Game Design: A Book of Lenses (3rd ed., 2019) · Mirza-Babaei et al., biometric storyboards in games user research (CHI 2013)

CASE STUDY · LEFT 4 DEAD · VALVE · 2008

Left 4 Dead’s AI Director



WHAT THEY DID

The Director estimates each player’s stress and cycles through four states: build up, sustain peak, peak fade and relax. It spawns threats in build up and holds them back while players recover.

WHAT IT TEACHES

A pacing curve can be written as rules and run live, so every session gets its own curve.

DISCUSS

Which parts of your pacing should be authored, and which should adapt to the player?

SOURCE Booth, The AI Systems of Left 4 Dead (Valve, 2009)

CASE STUDY · JOURNEY · THATGAMECOMPANY · 2012

Journey’s emotional arc

WHAT THEY DID

Jenova Chen charted the intended emotion across the whole game, combining three-act structure with the hero’s journey. The team shaped the world’s terrain from that curve.

WHAT IT TEACHES

An intensity curve can drive the geography of a level, not only the timing of encounters.

DISCUSS

Could the layout of your level follow your intensity curve?



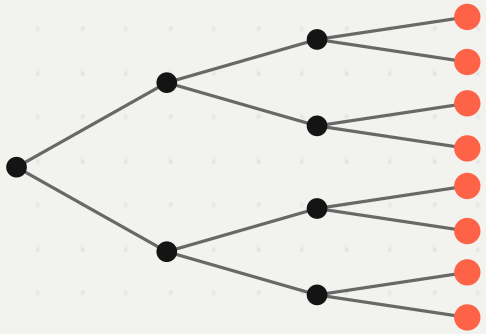
Jenova Chen at Game Developers Conference China, 2011.

Official GDC · CC BY 2.0 · Wikimedia Commons

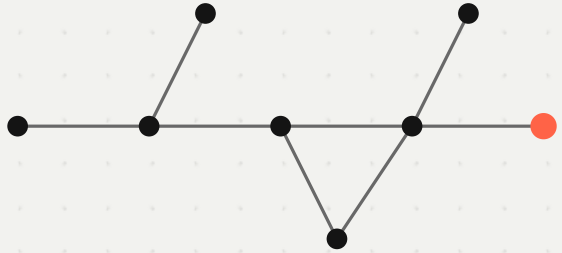
SOURCE Sheffield, The journey to create Journey: the quest for emotion (Game Developer, 2013) · Chen, Designing Journey (GDC 2013)

Branching patterns

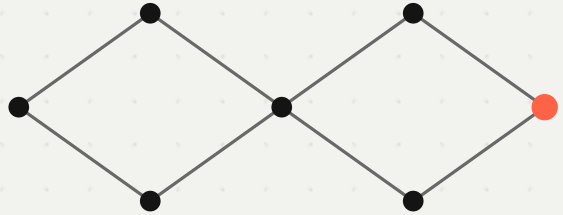
Time cave



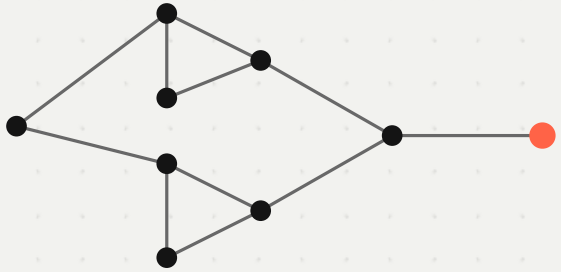
Gauntlet



Branch and bottleneck



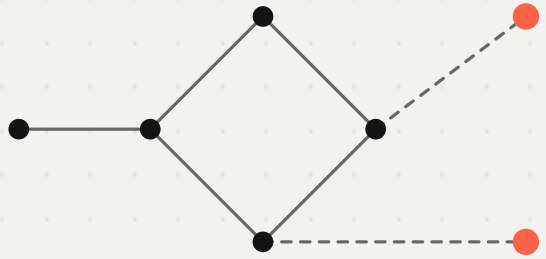
Quest



Sorting hat



Loop and grow

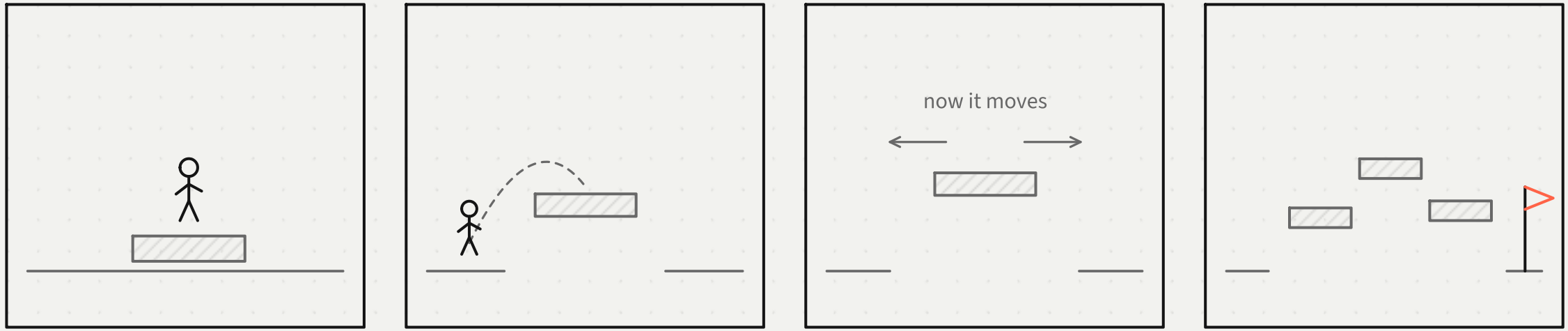


Every branch multiplies content and state. Standard patterns keep that under control.

Beyond branching: storylets unlock small scenes by state (qualities), instead of following a fixed tree.

SOURCE Ashwell, Standard Patterns in Choice-Based Games (2015) · Short, Beyond Branching: Quality-Based, Salience-Based, and Waypoint Narrative Structures (2016)

Four-step levels



1 · Introduce
ki · safely

2 · Develop
shō · add risk

3 · Twist
ten · change the rule

4 · Conclude
ketsu · test it all

Hayashida’s structure for Super Mario 3D Land, after kishōtenketsu. A level board is a lesson plan.

SOURCE Hayashida on four-step level design for Super Mario 3D Land (2012 interview)

CASE STUDY · SUPER MARIO BROS. · NINTENDO · 1985

Levels on graph paper

WHAT THEY DID

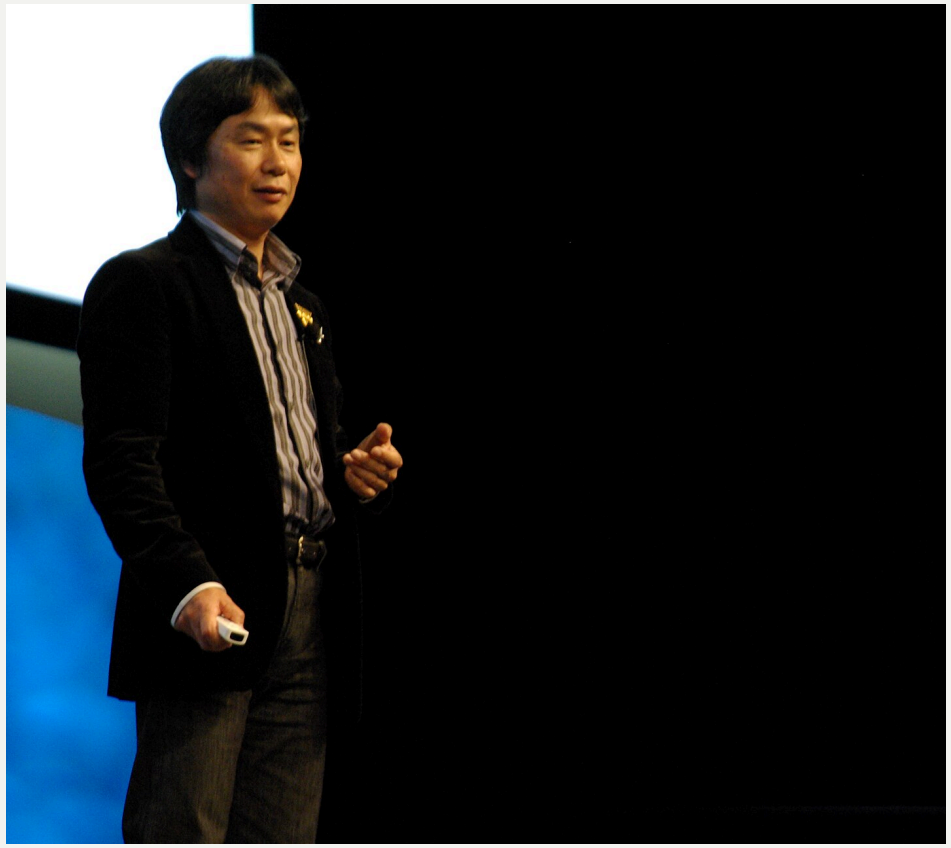
Miyamoto and Tezuka drew levels block by block on graph paper, edited them with tracing paper laid on top, and handed the sheets to the programmers.

WHAT IT TEACHES

A level board is a working document between design and engineering, made to be edited.

DISCUSS

What is your tracing paper: the cheapest way to edit your board?

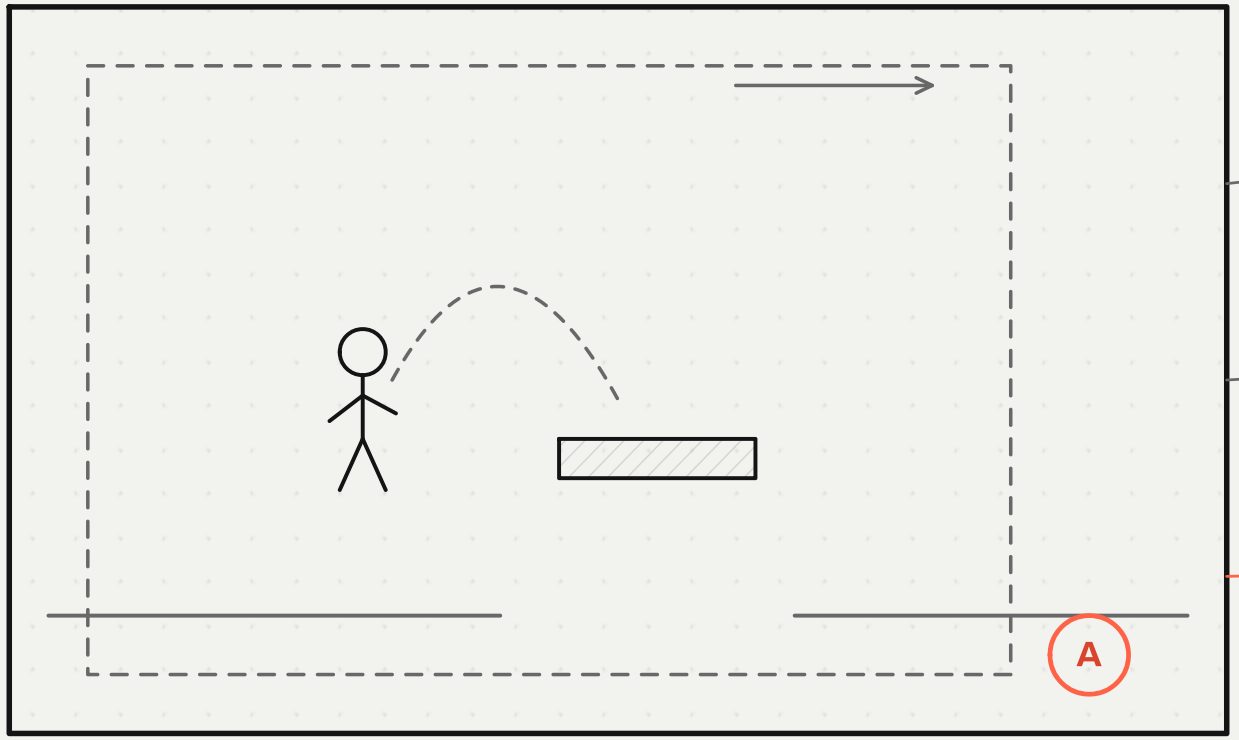


Shigeru Miyamoto speaking at GDC 2007.

Vincent Diamante · CC BY-SA 2.0 · Wikimedia Commons

SOURCE These sheets of graph paper were used to design Super Mario Bros. (ArchDaily)

Annotate the 3Cs



Character
can jump, and jump again in the air

Camera
side view, leads to the right

Controls
A to jump · A again in the air

Without the 3Cs, a gameplay panel is just a cutscene still.

STANDARD

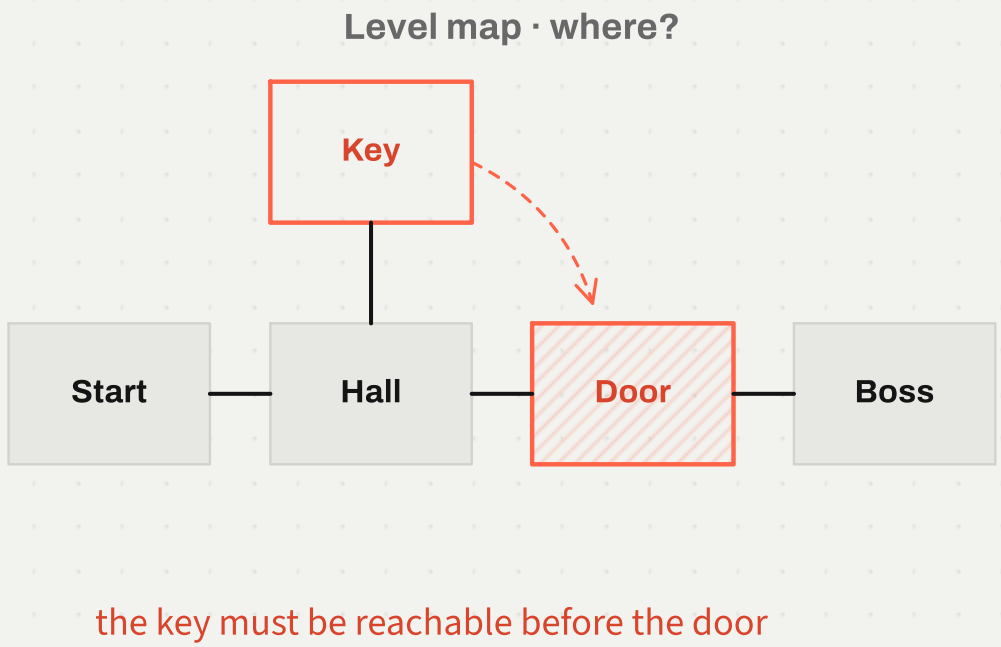
Annotating a gameplay panel

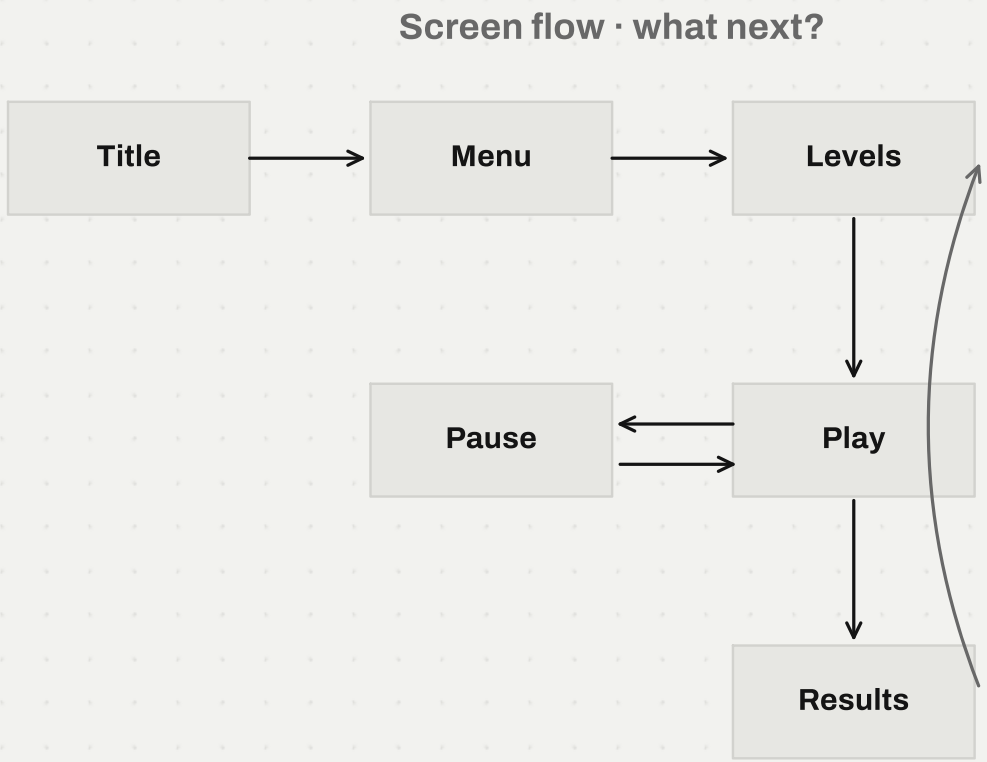
Character	What the player character can do here
Camera	What the player sees, and who controls the view
Controls	The inputs, as button glyphs
Goal	What the player is trying to do
State	Health, inventory, flags that matter here
Outcomes	What happens on success, and on failure
Teaching intent	What this panel teaches, if anything

Beat chart: common rows

Level and location	Where the player is
Story beat	What changes in the story
New mechanic	What the player learns or unlocks
Threats	Enemies and hazards
Rewards	Pickups, upgrades, secrets
Intensity	Planned level, low to high
Music and mood	What it should sound and feel like
Play time	Estimated minutes

Rooms and screens are boards too





“Where?” needs a map. “What next?” needs a flow.

SOURCE Totten, An Architectural Approach to Level Design · Hodent, The Gamer’s Brain (2017)

In the studio: where each board lives

PHASE	GOAL	BOARDS IN USE	ENDS AT
Concept	Is it worth making?	One-sheet, ten-pager, mood boards, key art	Greenlight
Pre-production	Find the fun and prove it can be built	Beat charts, paper prototypes, greybox levels, gameplay panels, first animatics	Vertical slice
Production	Build all the content	Cinematic boards → animatic → previz; a map and beat chart per level	Alpha: feature complete
Polish and test	Make it work and feel right	Playtest reports, measured intensity curves, bug lists	Beta: content complete → Gold: ship
Live	Keep it running and growing	Beat charts for events and seasons, analytics	Each update

Phase names and gates vary by studio. The order is common.

SOURCE Chandler, The Game Production Toolbox (2020) · Lemarchand, A Playful Production Process (2021)

Who makes the boards, and who reads them

ROLE	MAKES	USES THE BOARDS TO
Design	Beat charts, level maps, gameplay panels	Checks every board against the pillars
Art	Concept art, cinematic boards, key frames	Takes mood, scale and staging from beat charts and maps
Engineering	Prototypes and greybox tools	Turns 3C annotations into controls and camera; flags cost
Audio	Temp sound for animatics	Places music and sound cues on the intensity curve
UX	Screen flows and wireframes	Checks readability and onboarding in panels
QA	Test plans	Turns beat charts and flows into what to test
Production	Schedules and milestones	Reads the stack for scope, dependencies and risk; runs reviews

SOURCE Chandler, The Game Production Toolbox (2020)

How studios communicate design

One-sheet

One page that sells the game: genre, platform, audience, what makes it different.

Ten-pager

The whole game in about ten pages, for the team and for publishers.

One-page design

One annotated diagram per system or level, posted where the team can see it.

Design pillars

Three to five short statements of the intended experience. Decisions are checked against them.

Beat chart

The pacing plan across levels: mechanics, threats, story beats, intensity.

Animatic review

Timed boards reviewed by leads before expensive production starts.

Playtest report

What players did and felt, compared with what was planned.

Postmortem

What went right and wrong, written after shipping and often shared publicly.

Short and visual beats long and complete.

SOURCE Rogers, Level Up! The Guide to Great Video Game Design (3rd ed.) · Librande, One-Page Designs (GDC 2010)

Working norms

Find the fun first

Prove the core loop in a cheap prototype before scaling up.

Greybox before art

Block levels out in plain shapes. Art arrives once the space plays well.

Vertical slice

A short section at final quality that proves both the game and the pipeline.

Candid critique

Review the work, not the person. Notes are specific and actionable.

Short over long

A one-page diagram beats a long document nobody reads.

Cut to the pillars

If a feature or scene does not serve the pillars, it goes.

SOURCE Catmull with Wallace, Creativity, Inc. (2014) · Fullerton, Game Design Workshop

STANDARD

Playtest protocol

One question	Decide what this session must answer before it starts.
Right players	Recruit people who match the audience, not only the team.
Watch, don’t help	Let players struggle; note where they stop, look and fail.
Think aloud	Ask players to say what they are thinking as they play.
Ask after	Hold questions until the end, so you don’t steer play.
Observe, then interpret	Write what happened before what it means.
Design as hypothesis	Treat the design as a hypothesis and the playtest as an experiment (Valve).

Critique protocol

The work, not the person	Notes describe the board, never its author.
See, then think	Say what you see before what you think it means.
Tie it to the goal	Connect each note to the pillars or the intended feeling.
Suggest, don’t prescribe	Offer options; the author decides.
A simple format	“I noticed…”, “I wonder…”, “What if…”

Myth vs reality

MYTH

~~Game storyboards are cutscene storyboards.~~

REALITY

Cinematics are one layer of nine. Most game boards are charts, maps, graphs and annotated panels.

MYTH

~~More branches means more agency.~~

REALITY

Agency comes from consequences the player can feel. Branches multiply cost; bottlenecks and storylets keep it manageable.

TRY THIS · 20 MIN

Board World 1-1

1. Draw the first screen as a gameplay panel.
2. Annotate the 3Cs and the teaching intent.
3. Sketch the intensity curve for the first 30 seconds.

OUTPUT

One annotated panel and one curve.

TRY THIS · 15 MIN

Same story, two shapes

1. Take a scene with three choices in a row.
2. Draw it as a time cave, then as branch-and-bottleneck.
3. Count the scenes and endings each one needs.

OUTPUT

Two graphs and the counts.

INQUIRY

You are boarding a stealth level the player can enter from three sides.

Q1 Which layer do you board first: the map, the beat chart or the panels?

Q2 How do you show a guard’s sightline and the player’s choice on one panel?

Q3 What does the beat chart look like if the player is never seen?

Check yourself

1. **Name three units a game storyboard can use besides the shot.**

   Beat, choice node, room, gameplay panel with input, screen, intensity over time.

2. **What problem do branching patterns solve?**

   Content and state multiply with every branch. Patterns keep the structure coherent and affordable.

3. **What do the 3Cs add to a gameplay panel?**

   What the character can do, what the camera shows and what the player presses: the interaction, not just the picture.

4. **What is a vertical slice for?**

   A short section at final quality that proves the game is fun and that the team’s pipeline can produce it, before full production.

EX

Exemplars: what to steal

What have the best teams done with boards, graphs, worlds and characters?

Steal the structure, not the surface.

BRANCHING, MADE VISIBLE

Detroit: Become Human

QUANTIC DREAM · 2018

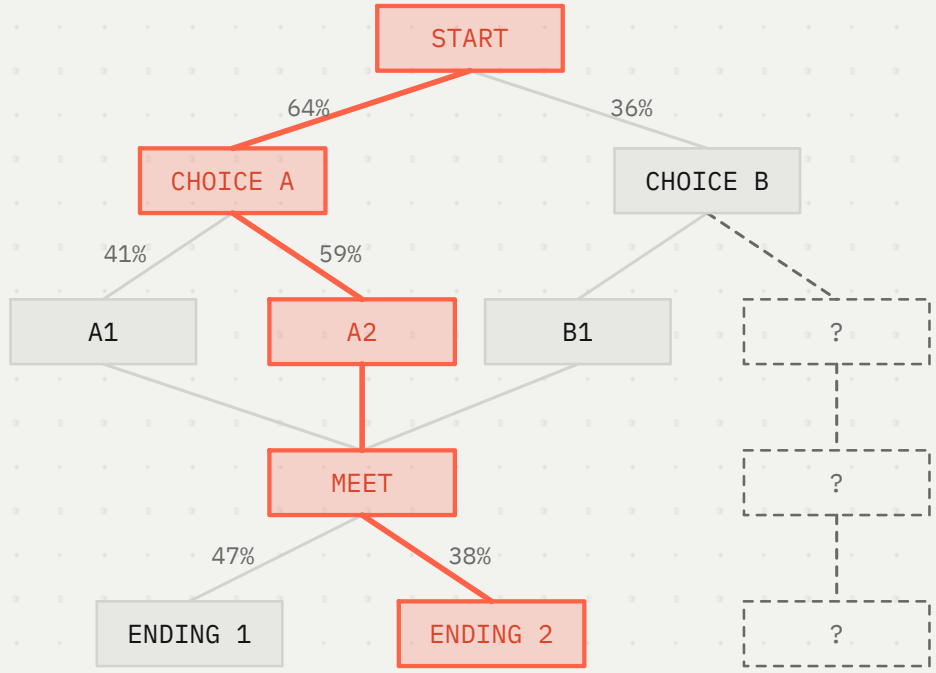
WHAT THEY DID

At the end of every chapter the game shows its flowchart: the paths you took, the ones you missed, and what percentage of players worldwide chose each option.

STEAL THIS

Keep a branch graph for every chapter, and treat it as a design tool. If you would be proud to show it to players, the structure is probably sound.

- M Choices that lock and unlock paths
- D Replays, comparison with other players
- A Responsibility, curiosity



ILLUSTRATIVE CHAPTER FLOWCHART · % = SHARE OF PLAYERS · DASHED = NOT YET FOUND

SOURCE Detroit: Become Human (Wikipedia)

KNOWLEDGE AS PROGRESSION

Outer Wilds

MOBIUS DIGITAL · 2019

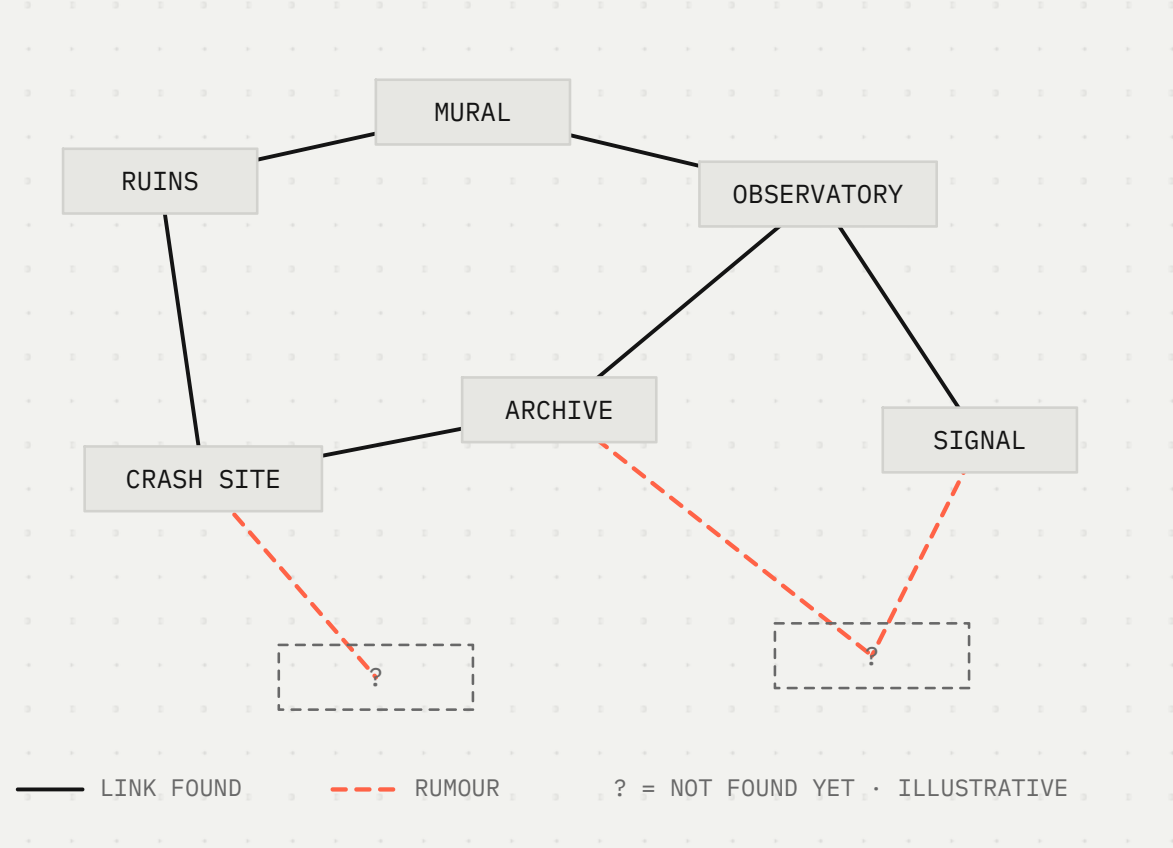
WHAT THEY DID

Nothing unlocks except what you know. The ship log has a Rumor mode that draws your discoveries as a graph, linked by the rumours that connect them; its layout depends on the order you found things.

STEAL THIS

Board what the player knows, not only where they are. A knowledge graph can be the level design.

- M Clues, a time loop, a log that records them
- D Following leads, forming theories
- A Discovery



SOURCE Outer Wilds ship log: Map and Rumor modes (Outer Wilds wiki)

STORY TOLD THROUGH A LOOP

Hades

SUPERGIANT GAMES · 2020

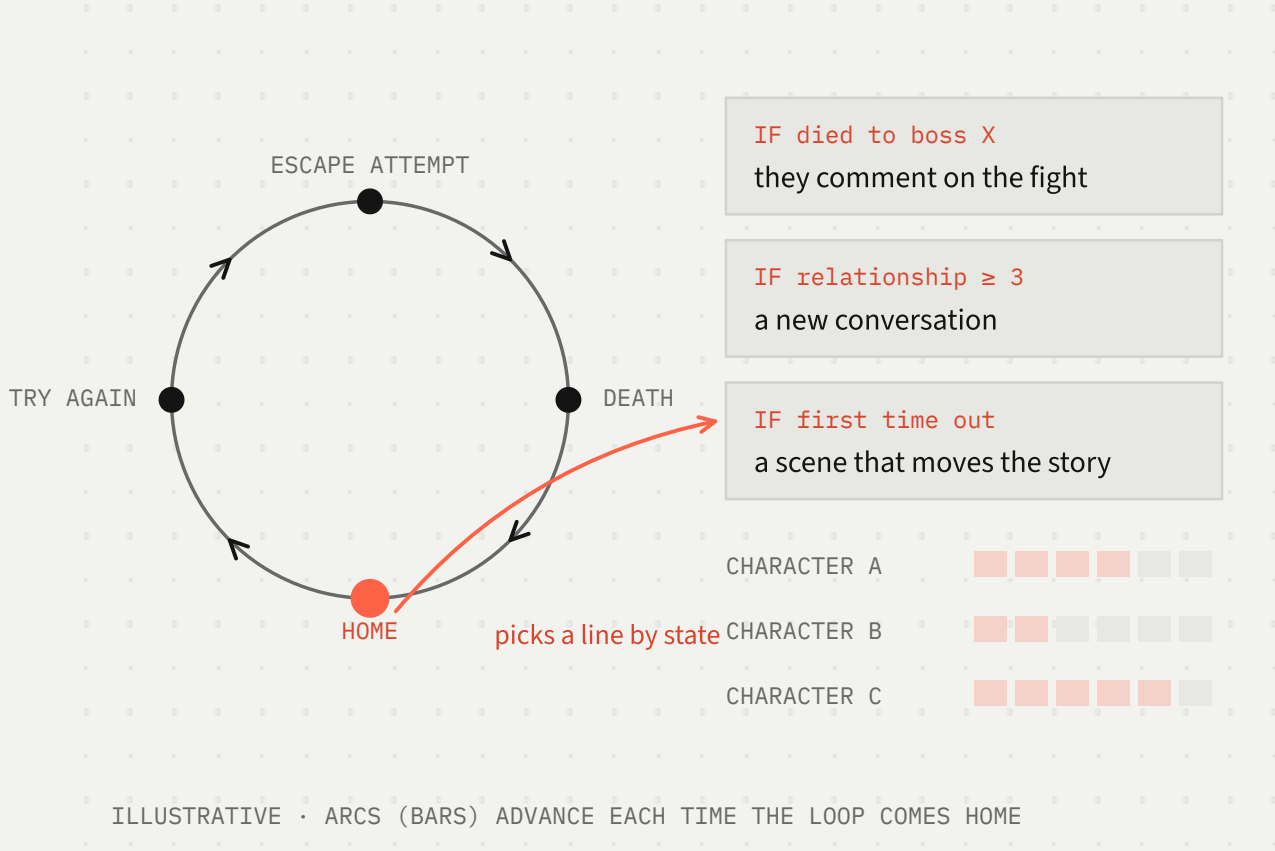
WHAT THEY DID

Every death sends you home, where characters react to what just happened. More than 22,000 lines of dialogue are chosen by the state of your runs, so failing moves the story forward.

STEAL THIS

Attach arcs to your loop. Write scenes as small pieces unlocked by state, so repetition feels like progress.

- M Runs, death, a home base, state-keyed dialogue
- D Retrying with something new to hear
- A Narrative, challenge, fellowship



ILLUSTRATIVE · ARCS (BARS) ADVANCE EACH TIME THE LOOP COMES HOME

SOURCE Kasavin & Korb, Breathing Life into Greek Myth: The Dialogue of Hades (GDC 2021) · Game Developer, How Supergiant weaves narrative rewards into Hades' cycle of perpetual death

WORLD BUILDING WITH TERRAIN

The Legend of Zelda: Breath of the Wild

NINTENDO · 2017

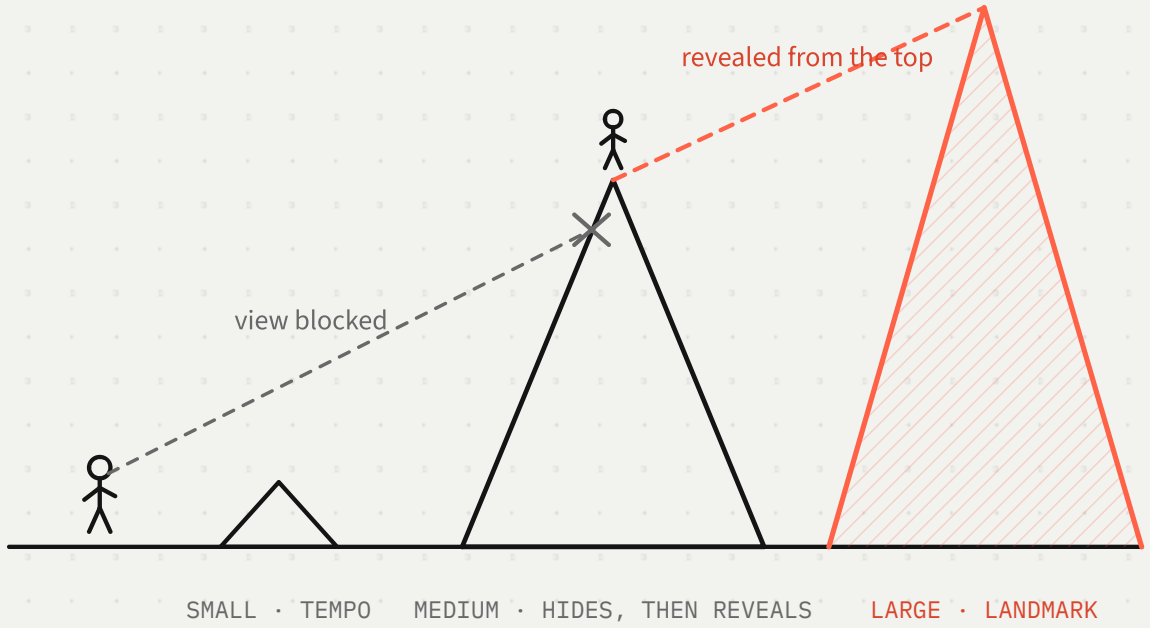
WHAT THEY DID

The field team described a “triangle rule”: large triangles are landmarks, medium ones hide what lies behind them, small ones set the tempo. Triangles give a choice (over or around) and make players wonder what is on the other side.

STEAL THIS

Board sightlines. Decide what the player sees, what is hidden and when it is revealed.

- M Climbable terrain shaped as triangles
- D Detours, climbs, sudden reveals
- A Discovery



SIDE VIEW · AFTER NINTENDO’S CEDEC 2017 TALK

SOURCE Fujibayashi & Yonezu on the field design of Breath of the Wild (CEDEC 2017), via Game Developer

CAMERA AS A DESIGN DECISION

God of War

SANTA MONICA STUDIO · 2018

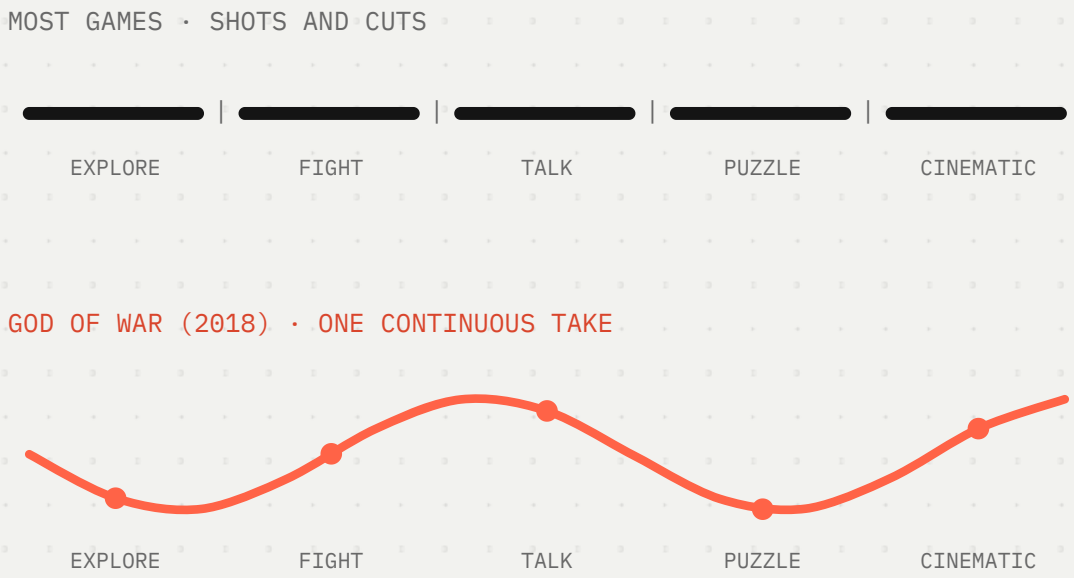
WHAT THEY DID

The whole game is one continuous shot with no camera cuts. Director Cory Barlog wanted to keep players with Kratos and Atreus at every moment; it made transitions much harder to plan and extended production.

STEAL THIS

Choose your camera grammar before you board. It changes every panel, transition and cinematic you draw.

- **M** A camera that never cuts
- **D** Fights, talks and travel flowing together
- **A** Intimacy with the characters



No cuts and no loading screens: every transition has to be designed.

SOURCE Variety, Creating God of War's single-shot cinematography (2018)

CHARACTER THROUGH MECHANICS

What Remains of Edith Finch

GIANT SPARROW · 2017

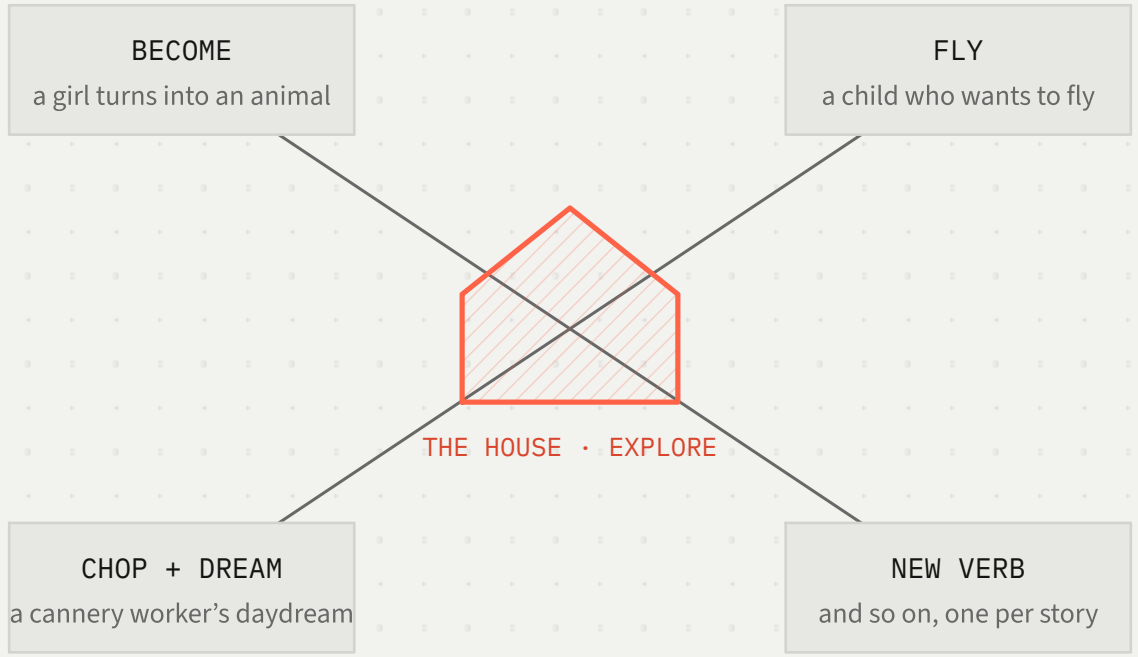
WHAT THEY DID

You explore a family house, and each family member's story plays with a different mechanic. In one, a cannery worker chops fish heads while his daydream of a kingdom slowly takes over the screen.

STEAL THIS

Give each character a verb. A hub with spokes lets every story have its own rules.

- M A different mechanic for each story
- D Learning a new way to play every few minutes
- A Narrative, discovery, empathy



HUB AND SPOKES · EACH STORY HAS ITS OWN RULES

SOURCE What Remains of Edith Finch (Wikipedia)

STATS AS CHARACTERS

Disco Elysium

ZA/UM · 2019

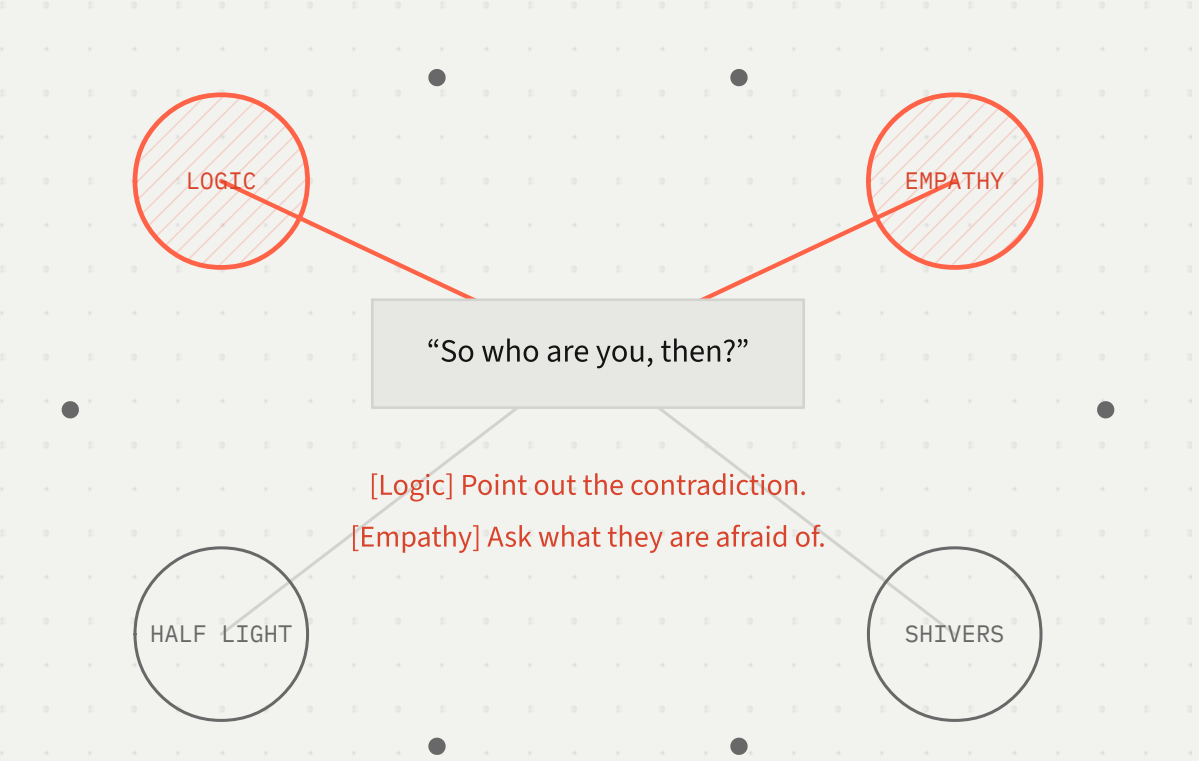
WHAT THEY DID

The detective's 24 skills speak as voices in his head. Logic, Empathy, Half Light and others interrupt conversations with their own views, and the points you invest decide which voices you hear.

STEAL THIS

Turn systems into characters. A stat that talks is a stat players care about.

- M 24 skills grouped under four attributes
- D Inner arguments that open dialogue options
- A Expression, narrative



24 SKILLS · 4 ATTRIBUTES · PASSED CHECKS OPEN OPTIONS · ILLUSTRATIVE

SOURCE Disco Elysium (Wikipedia)

MECHANICS CARRY THE THEME

Celeste

MADDY MAKES GAMES · 2018

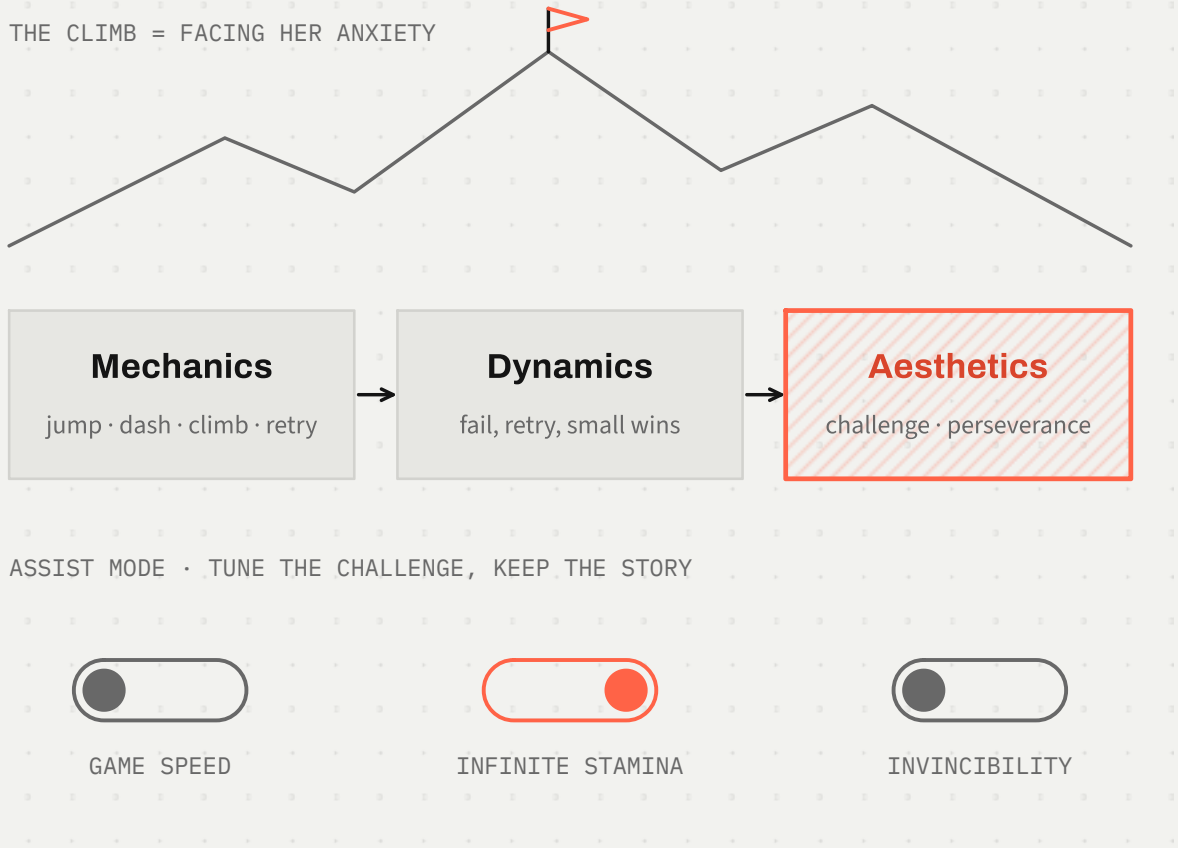
WHAT THEY DID

Madeline climbs a mountain while dealing with anxiety and depression; her doubts appear as a dark copy of herself. Assist Mode lets players slow the game, add stamina or turn on invincibility, because players come in at different skill levels.

STEAL THIS

Align the aesthetic with the character's arc, and let players tune the challenge without changing the story.

- **M** Jump, dash, climb with stamina, instant retry
- **D** Fail, retry, small wins
- **A** Challenge, perseverance



SOURCE Celeste (Wikipedia)

SMALL RULES, HEAVY FEELINGS

Papers, Please

LUCAS POPE · 2013

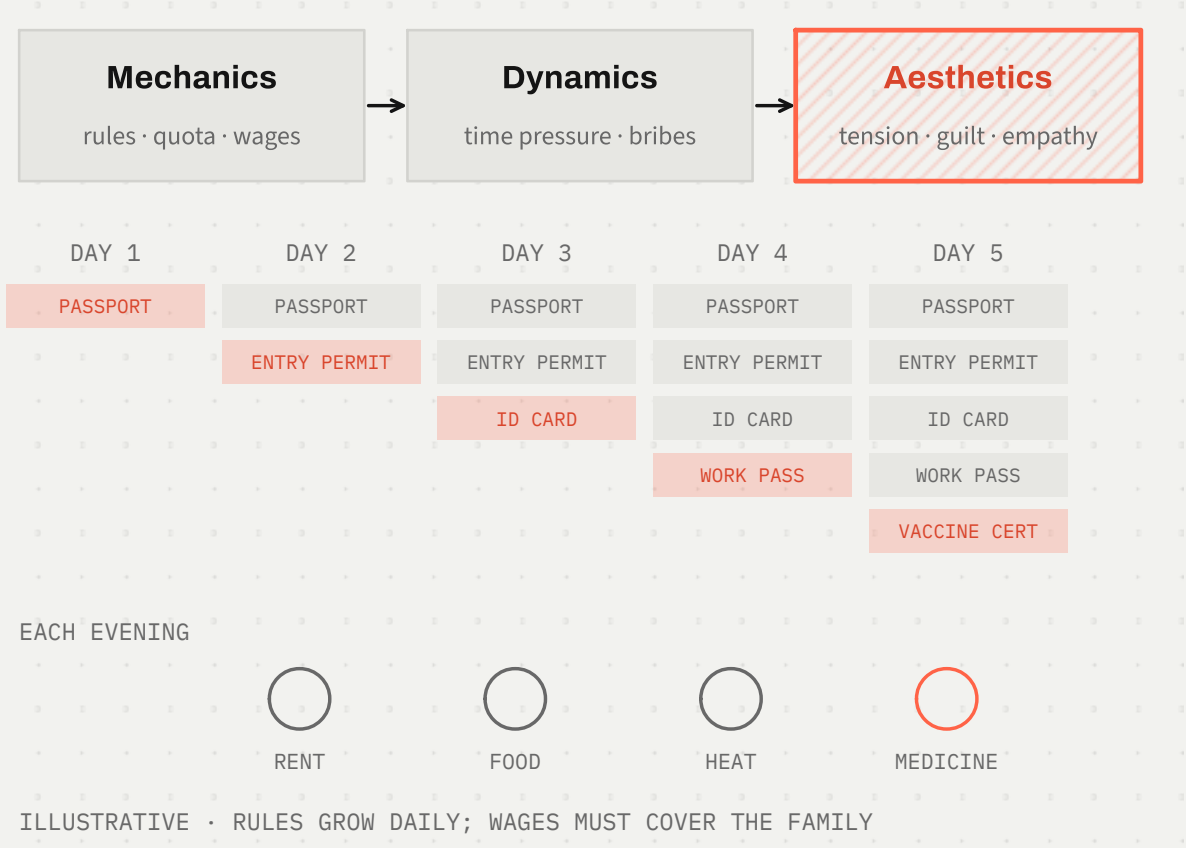
WHAT THEY DID

You are a border inspector checking documents against rules that grow every day, paid per person processed, with a family to feed. Bribes and desperate stories arrive at the booth.

STEAL THIS

Pair a strict rule system with human stakes. The moral weight comes from the mechanics, not from cutscenes.

- M Document rules, quotas, wages, family costs
- D Time pressure, bending rules, bribes
- A Tension, guilt, empathy



SOURCE Papers, Please (Wikipedia)

What to steal

GAME	STRUCTURE	STEAL THIS
Detroit: Become Human	Branching, made visible	Keep a branch graph for every chapter, and treat it as a design tool.
Outer Wilds	Knowledge as progression	Board what the player knows, not only where they are.
Hades	Story told through a loop	Attach arcs to your loop.
The Legend of Zelda: Breath of the Wild	World building with terrain	Board sightlines.
God of War	Camera as a design decision	Choose your camera grammar before you board.
What Remains of Edith Finch	Character through mechanics	Give each character a verb.
Disco Elysium	Stats as characters	Turn systems into characters.
Celeste	Mechanics carry the theme	Align the aesthetic with the character’s arc, and let players tune the challenge without changing the story..
Papers, Please	Small rules, heavy feelings	Pair a strict rule system with human stakes.

Steal the structure, not the surface.

04

How AI helps

Which layers can AI take on, and which need a human?

AI can generate, check and even play the board.
Keeping it consistent and judging it are the hard parts.



Server racks in a data centre.
BalticServers.com · CC BY-SA 3.0 · Wikimedia Commons

One stack, many models

LAYER	UNIT	REPRESENTATION	KIND OF MODEL
Concept	The pitch	Text	Language models
Narrative structure	The beat	Text	Language models
Branching structure	The choice	Graph	Language models + explicit state
Experience pacing	Intensity over time	Curve	Critics and simulated players
Level and space	The room	Map	Procedural generation, 3D generation
Gameplay moment	The panel + input	Image	Image models with conditioning
Cinematic	The shot	Video	Video models
Interface flow	The screen	Flow	Interface generation
Validation	The playtest	Play	World models, agent playtesters

The hard part is keeping all of these consistent with each other.

SOURCE Gallotta et al., Large Language Models and Games: A Survey and Roadmap (2024)

Where AI enters the pipeline

STAGE	TRADITIONALLY	AI'S ROLE	THE HARD ENGINEERING PROBLEM
Premise → beat sheet	Writer and designer	Language models generate beats and branching quests	**Keeping branches coherent and tracking state**
Beats → thumbnails	Concept artist	Image models with sketch, pose and reference conditioning	**The same character, style and world in every frame**
Thumbnails → animatic	Editor	Video generation models	**Controlling camera and timing**
Animatic → previz	Level designer	3D generation and playable world models	**Turning frames into something interactive**
Review	Director and playtesters	Language or vision models as critics	**Evaluation: what makes a storyboard good?**

Four hard problems

Consistency

Same character, style, world and story state across frames, branches and layers.

Control

The framing, camera and timing you intend, not what the model prefers.

Branching coherence

Every branch tracks state, and no path contradicts another.

Evaluation

Measuring whether a board is good, and checking the AI critic itself.

SOURCE Zhou et al., StoryDiffusion: Consistent Self-Attention for Long-Range Image and Video Generation (2024) · Zhang, Rao & Agrawala, Adding Conditional Control to Text-to-Image Diffusion Models (ControlNet, 2023) · Riedl & Bulitko, Interactive Narrative: An Intelligent Systems Approach (AI Magazine, 2013) · Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025)

What creatives need

From interviews with game creatives behind Microsoft’s Muse (Kanervisto et al., 2025).

Consistency

Generated play stays true to the game’s rules and physics.

Diversity

The tool offers genuinely different options, not variations on one.

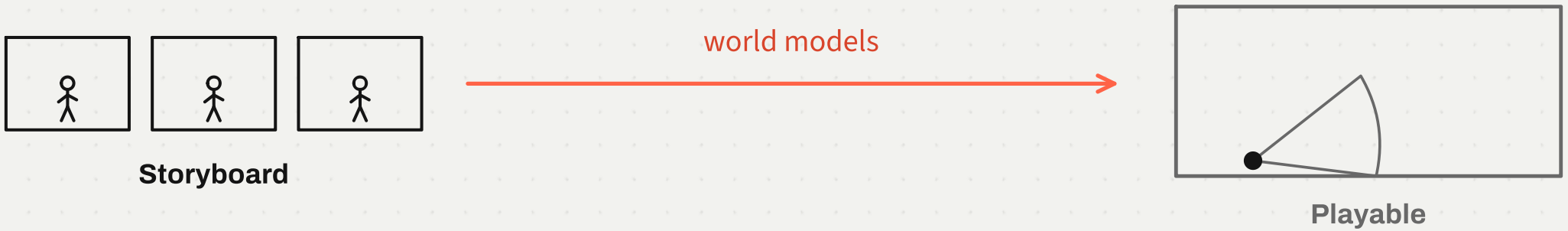
Persistency

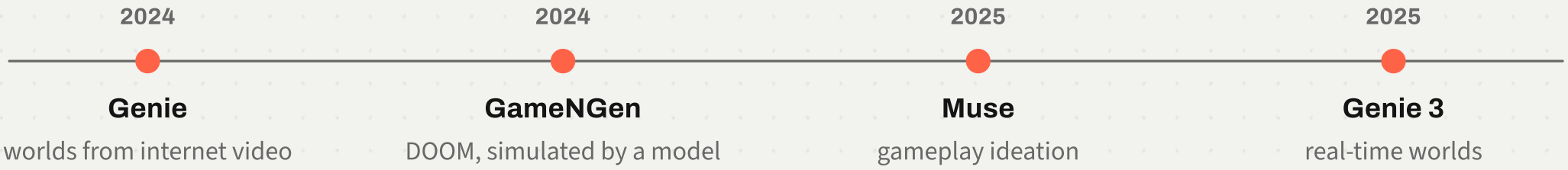
When a creator changes something, the change sticks in what is generated next.

Use these as evaluation criteria for any AI storyboard tool.

SOURCE Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025)

When the board becomes playable





If you can play the board, is it still a planning tool, or a tool for aligning a team's intent?

SOURCE Bruce et al., Genie: Generative Interactive Environments (2024) · Valevski et al., Diffusion Models Are Real-Time Game Engines (GameNGen, 2024) · Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025) · Google DeepMind, Genie 3 (2025)

CASE STUDY · RESEARCH MODEL · GOOGLE DEEPMIND · 2024

Genie 2: concept art you can walk into

WHAT THEY DID

Genie 2 turns a single image, including concept art, into an environment that a person or an agent can move through with keyboard and mouse. DeepMind presented it as a way for artists and designers to prototype quickly.

WHAT IT TEACHES

The jump from board to playable can now start from one frame. This is the board-to-prototype collapse in practice.

DISCUSS

If one concept painting becomes playable, what does the storyboard still need to specify?

SOURCE Google DeepMind, Genie 2: a large-scale foundation world model (2024)

CASE STUDY · BLEEDING EDGE · MICROSOFT RESEARCH AND NINJA THEORY · 2025

Muse: a model of one game

WHAT THEY DID

Muse was trained only on Bleeding Edge: over a billion frames and controller actions, about seven years of play. It continues gameplay from a short input, and creatives can edit what it generates. The weights and a demonstrator were released.

WHAT IT TEACHES

A world model can be a gameplay sketchpad, but it only knows the game it was trained on.

DISCUSS

Would a model trained on your last game help or limit your next one?



An Xbox One controller. Muse models what the screen shows and what the controller does.

Evan-Amos · Public domain · Wikimedia Commons

SOURCE Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025) · Xbox Wire, Empowering creators and players with Muse (2025)

CASE STUDY · ROBLOX · ROBLOX · 2025–26

Roblox: objects that come with behaviour

WHAT THEY DID

Cube turns a text prompt into a 3D mesh, and Roblox open-sourced the 3D model. In 2026 it added 4D generation: objects built from schemas, such as a five-part car, with behaviour attached so they drive or fly. Roblox reports players generated over 160,000 objects in one game's early access.

WHAT IT TEACHES

The generated unit is moving from a picture to a playable object. In 4D generation the asset carries its own mechanics.

DISCUSS

Roblox reported 64% more play time among players who used it. Is play time the right measure of a design tool?

SOURCE Roblox, Introducing Roblox Cube: our core generative AI system for 3D and 4D (2025) · Roblox, Accelerating creation, powered by Roblox's Cube foundation model (2026)

CASE STUDY · OPEN-WORLD GAMES · UBISOFT LA FORGE · 2023

Ghostwriter: drafts for writers

WHAT THEY DID

Ghostwriter generates first drafts of barks, the short lines NPCs say when something happens. Writers choose and polish drafts instead of writing them from scratch. The announcement also drew criticism about writers' jobs.

WHAT IT TEACHES

AI in production tends to land first on high-volume, low-stakes content, and it raises authorship questions on the way.

DISCUSS

Which writing in your game would you hand to a model, and which never?

SOURCE Ubisoft, The Convergence of AI and Creativity: Introducing Ghostwriter (2023)

AI in studios today: what is claimed

COMPANY	TOOL	LAYER	WHAT IT SPEEDS UP	EVIDENCE
Roblox	Cube (2025)	Assets for level and space	Text to a usable 3D mesh in seconds, instead of hours of modelling	**Company claim**
Roblox	4D generation (2026)	Objects with behaviour	160,000+ objects generated; 64% more play time among players who used it	**Company metric**
Google DeepMind	Genie 2 (2024)	Playable previz	Concept art turned into a playable environment	**Research demo**
Microsoft, Ninja Theory	Muse (2025)	Gameplay ideation	Generated gameplay that creatives can edit	**Peer-reviewed**
Ubisoft	Ghostwriter (2023)	Narrative: barks	First drafts that writers choose and polish	**Company description**
Ubisoft	Teammates (2025)	Play-time NPCs	Squadmates that respond to the player's voice; closed playtest	**Prototype**
Embark Studios	The Finals (2023)	Audio: voice lines	Text-to-speech built on contracted actors: lines in hours, not months	**Developer statement**
Square Enix	QA automation	Validation	Generative AI to handle 70% of QA and debugging by the end of 2027	**Target, not result**

Most numbers come from the companies themselves. Ask what was measured: speed, volume, play time or quality?

SOURCE Roblox, Introducing Roblox Cube: our core generative AI system for 3D and 4D (2025) · Roblox, Accelerating creation, powered by Roblox's Cube foundation model (2026) · Google DeepMind, Genie 2: a large-scale foundation world model (2024) · Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025) · Ubisoft, The Convergence of AI and Creativity: Introducing Ghostwriter (2023) · Ubisoft, Ubisoft reveals Teammates (2025) · Game Developer, Embark Studios: AI let devs "do more with less" when making The Finals (2023) · VGC, Square Enix wants generative AI doing 70% of its QA and debugging by the end of 2027 (2025)

Four ways studios use AI to go faster

Skip the blank page

Drafts to react to instead of starting from nothing.

Ghostwriter, Cube

Shorten idea to playable

Ideas you can try in minutes instead of weeks.

Genie 2, Muse, 4D generation

Automate volume

High-count, repetitive work: lines, barks, test passes.

The Finals, Square Enix QA

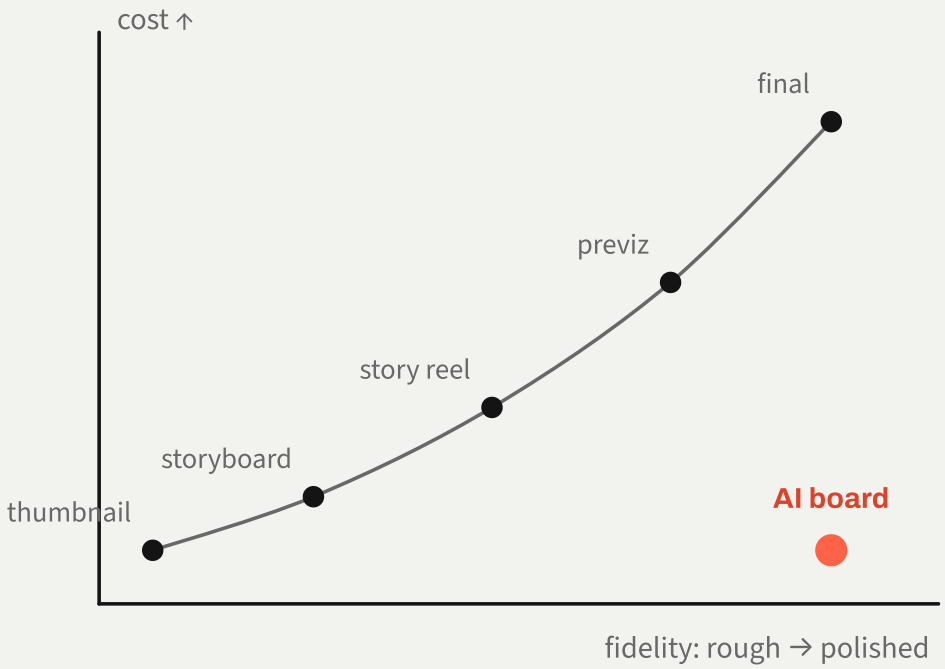
Hand creation to players

Players generate content inside the game.

Roblox 4D generation

Each one saves time somewhere and creates new work somewhere else: review, curation, evaluation.

Where AI may hurt



polished and cheap: will people still critique it?

Premature fidelity HYPOTHESIS

AI makes boards cheap and polished at the same time. Polished boards may attract less critique, which undoes “fail cheaply”.

Inferred from Buxton and Catmull

Flat pacing CLAIM TO TEST

Models produce content easily but have no felt sense of rhythm or tension.

Raised in our brainstorm

Sameness EVIDENCE

AI raises individual creativity but lowers the diversity across a group.

Doshi & Hauser, 2024 (short stories)

SOURCE Doshi & Hauser, Generative AI enhances individual creativity but reduces the collective diversity of novel content (Science Advances, 2024) · Buxton, Sketching User Experiences (2007) · Catmull with Wallace, Creativity, Inc. (2014)

Mixed initiative: who does what

AI proposes

- Generate options at every layer
- Check consistency across frames and branches
- Critique against named lenses
- Play the board as a simulated player

The human decides

- Intent: what the player should feel
- Taste: what to keep and what to cut
- Pacing judgement
- The final call

SOURCE Yannakakis, Liapis & Alexopoulos, Mixed-Initiative Co-Creativity (FDG 2014) · Liapis, Yannakakis & Togelius, Sentient Sketchbook (2013)

Myth vs reality

MYTH

~~Better image models will make better storyboards.~~

REALITY

Image quality is the easy part. Consistency, control and evaluation are the hard ones.

MYTH

~~AI will replace storyboard artists.~~

REALITY

Current tools draft; people choose, direct and judge. How roles shift is an open question, not a settled one.

TRY THIS · 20 MIN

Drift test

1. Generate the same character in five panels with an image model.
2. List what drifts: face, outfit, proportions, layout.
3. Add one conditioning input (a sketch or reference) and compare.

OUTPUT

A drift table, before and after.

TRY THIS · 20 MIN

Critic test

1. Plant three errors in a board: one continuity, one state, one pacing.
2. Ask an AI critic to review the board.
3. Score each error: caught, missed or made up.

OUTPUT

A three-by-three tally.

INQUIRY

A tool turns your beat sheet into a full animatic in five minutes.

Q1 What would you check before showing it to the team?

Q2 Who now owns the pacing decisions?

Q3 How would you test whether the team critiques it less than a rough board?

Check yourself

1. **Why is consistency the central engineering problem for AI storyboarding?**

   Each layer uses a different model and representation, and characters, style, world and story state must agree across frames, branches and layers.

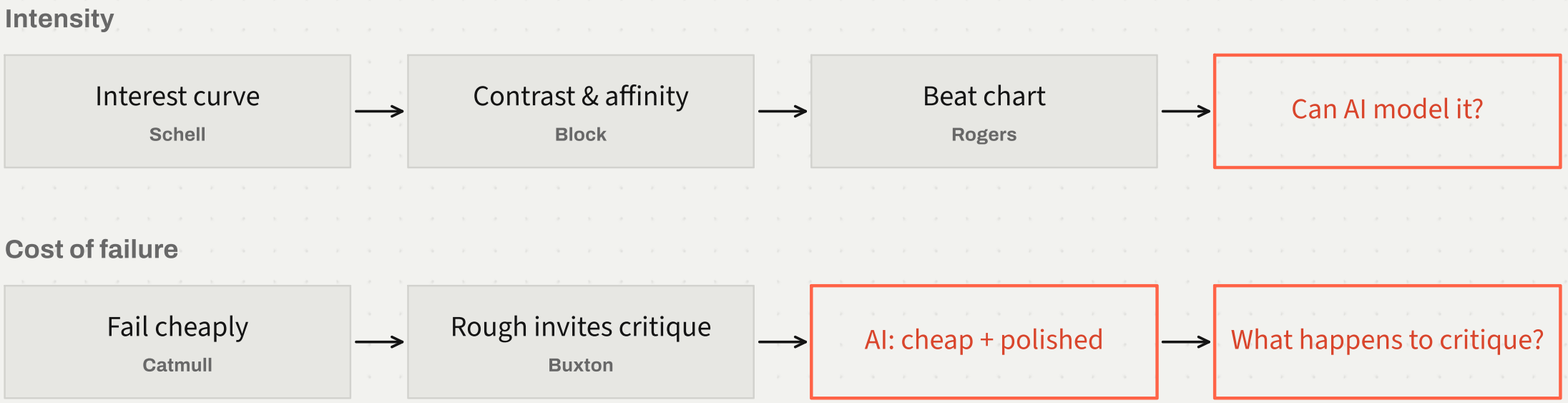
2. **What three needs did the Muse study identify for AI ideation tools?**

   Consistency, diversity and persistency.

3. **What is “premature fidelity”, and how strong is the evidence for it?**

   AI makes boards polished early, which may reduce critique and exploration. It is a hypothesis to test, not an established finding.

Two threads through all four modules



Four angles for the talk

A · Tool survey

What exists at each stage.

Useful, but dates quickly.

B · Engineering systems

Consistency, control, branching and evaluation as open problems.

Best fit for an AI engineering course: the spine.

C · Storyboard → prototype collapse

When the board is playable, does storyboarding disappear?

The provocative one: the hook.

D · Authorship and craft

What stays human: taste, intent, pacing judgement; artists' jobs and IP.

The closing question.

Questions to argue about

1. If AI can produce a full animatic in minutes, is the storyboard still a planning tool, or a communication tool for aligning a team's intent?

2. Which layer of the stack does AI make worse rather than better? (One guess: pacing.)

3. When the storyboard becomes playable, what is left for the storyboard to do?

A reading path

1. Hunicke, LeBlanc & Zubek, MDA: A Formal Approach to Game Design and Game Research (2004) COURSE
2. Schell, The Art of Game Design: A Book of Lenses (3rd ed., 2019) COURSE
3. McCloud, Understanding Comics (1993)
4. Block, The Visual Story
5. Catmull with Wallace, Creativity, Inc. (2014) COURSE
6. Jenkins, Game Design as Narrative Architecture (2004)
7. Rogers, Level Up! The Guide to Great Video Game Design (3rd ed.)
8. Ashwell, Standard Patterns in Choice-Based Games (2015)
9. Kanervisto et al., World and Human Action Models towards Gameplay Ideation (Nature, 2025)

The full list, with optional readings, is in the study guide.

NEXT

Distil the important ideas into an AI storyboarding demo.

Start from Module 4: each hard problem and each risk is a candidate experiment.

IMAGE CREDITS

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Charge-concept art-storyboard.png
Andy Goralczyk / Blender Foundation · CC BY 4.0

Ryzom storyboard 1.jpg
Ryzom · CC BY-SA 2.0

BalticServers data center.jpg
BalticServers.com · CC BY-SA 3.0

Toru Iwatani, creator of Pac-Man, at GDC 2011 (cropped to upper body).jpg
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Takashi Tezuka, Shigeru Miyamoto and Kōji Kondō.jpg
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Jenova Chen – Game Developers Conference China 2011 (1).jpg
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Ryzom storyboard 2.jpg
Ryzom · CC BY-SA 2.0

Spring-concept art-storyboard.png
David Revoy / Blender Foundation · CC BY 4.0

SuperTuxKart in-race (2018).png
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Minetest screenshot.png
Perttu Ahola, Vanessa Ezekowitz et al. · CC BY-SA 3.0

0 A.D. alpha 25 - playing as Spartans.jpg
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